

Dietary fructose as a risk factor for non-alcoholic fatty liver disease (NAFLD)

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Abstract Glucose is a major energy source for the entire body, while fructose metabolism occurs mainly in the liver. Fructose consumption has increased over the last decade globally and is suspected to contribute to the increased incidence of non-alcoholic fatty liver disease (NAFLD). NAFLD is a manifestation of metabolic syndrome affecting about one-third of the population worldwide and has progressive pathological potential for liver cirrhosis and cancer through non-alcoholic steatohepatitis (NASH). Here we have reviewed the possible contribution of fructose to the pathophysiology of NAFLD. We critically summarize the current findings about several regulators, and their potential mechanisms, that have been studied in humans and animal models in response to fructose exposure. A novel hypothesis on fructose-dependent perturbation of liver regeneration and metabolism is advanced. Fructose intake could affect inflammatory and metabolic processes, liver function, gut microbiota, and portal endotoxin influx. The role of the brain in controlling fructose ingestion and the subsequent development of NAFLD is highlighted. Although the importance for fructose (over)consumption for NAFLD in humans is still debated and comprehensive intervention studies are invited, understanding of how fructose intake can favor these pathological processes is crucial for the development

of appropriate noninvasive diagnostic and therapeutic approaches to detect and treat these metabolic effects. Still, lifestyle modification, to lessen the consumption of fructose-containing products, and physical exercise are major measures against NAFLD. Finally, promising drugs against fructose-induced insulin resistance and hepatic dysfunction that are emerging from studies in rodents are reviewed, but need further validation in human patients.

Keywords Insulin resistance · Inflammation · Metabolic syndrome (MetS) · Gut microbiota · Herbal medicine · Biomarkers · The brain · Ethanol · Humans · Oxidative stress · Liver regeneration · ATP

Abbreviations

CYP2E1	Cytochrome 450 2E1
ER	Endoplasmic reticulum
GLUT	Glucose transporter
HDL	High-density lipoprotein
HFCS	High-fructose corn syrup
HFD	High-fat diet
IR	Insulin resistance
IRS-1/2	Insulin receptor substrate 1/2
LCN2	Lipocalin 2
LDL	High-density lipoprotein
LPS	Lipopolysaccharide
Ltf	Lactoferrin
MC4R	Melanocortin receptor 4
MetS	Metabolic syndrome
NAFLD	Non-alcoholic fatty liver disease
NASH	Non-alcoholic steatohepatitis
PAI-1	Plasminogen activator inhibitor-1
ROS	Reactive oxygen species
T2D	Type 2 diabetes mellitus
TG	Triglycerides

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TLR4 Toll-like receptor 4
 TNF- α Tumor necrosis factor- α

Introduction

The liver is an essential organ for protein, fats, and carbohydrate metabolism as well as catabolism and excretion of toxins. Any functional impairment will therefore affect the whole organism and leads to significant morbidity and mortality. Non-alcoholic fatty liver disease (NAFLD) encompasses a broad spectrum of pathological conditions ranging from simple steatosis to non-alcoholic steatohepatitis (NASH), fibrosis, and cirrhosis, and may ultimately lead to hepatocellular carcinoma (Conlon et al. 2013). The prevalence of NAFLD is estimated to be 20–30% of the general population in Europe and the USA (Gorden et al. 2011; Sapp et al. 2014). With an increase in the availability of high-calorie food, the use of fructose as a sweetener (e.g., in beverages) has been implicated in the increasing prevalence of the NAFLD and metabolic syndrome (MetS) (Bray et al. 2004; Vilà et al. 2011; Aroor et al. 2015). Fructose, a monosaccharide, naturally exists in fruits and vegetables. Fructose has the highest relative sweetness over other sugars; therefore, it is exploited in soft drinks and other commercial products (Montonen et al. 2007), particularly in the form of high-fructose corn syrup (HFCS). Dietary fructose is considered a risk factor for NAFLD and obesity (Bray et al. 2004; Nakagawa et al. 2006). However, the causal relationship between fructose consumption and liver pathogenesis still needs to be elucidated.

The understanding of the molecular, biochemical and pathological changes that cause the early-stages of NAFLD is a prerequisite to improve treatment strategies. This review is focused on the physiological influences of fructose consumption that lead to NAFLD and its associated abnormalities, such as the MetS, insulin resistance (IR), and disruption of gut microbiota. Furthermore, we suggest a novel hypothesis on how fructose may hit the liver and have reviewed possible therapeutic options against fructose-induced NAFLD. It also highlights how the brain regulates fructose intake and its adverse impact on the brain. This study recommends that more efforts should be taken to raise public awareness about the pros and cons associated with fructose consumption.

Fructose and glucose: global consumption and metabolism

Monosaccharide uptake and metabolism

Fructose is absorbed by enterocytes in the small intestine via GLUT2 and GLUT5 transporters (Soleimani 2011; Ritze

et al. 2014) by both a partially active process and facilitated diffusion in rodents (Riby et al. 1993). D-Fructose inhibits its own transport, whereas the presence of glucose could facilitate transport of fructose in intestine and renal brush border membrane (Korieh and Crouzoulon 1991; Mate et al. 2001; Sloboda et al. 2014). Thereafter, fructose is delivered to the liver via the portal vein and taken up by GLUT2, GLUT5, and GLUT8 transporters on the surface of hepatocytes (Sakar et al. 2009; Soleimani 2011; Narasimhan et al. 2015). Hepatocytes of GLUT8-lacking mice failed to take up fructose and subsequently had an attenuated fructose-induced steatosis (DeBosch et al. 2014).

Although both glucose and fructose are metabolized through the glycolytic pathway, the rate-limiting enzyme (phosphofructokinase) is bypassed and the initial steps differ in fructose metabolism from that of glucose (Cha et al. 2008). When zebra fish were treated with 4% (w/v) fructose or glucose, only the fructose-treated larvae had developed NASH: abnormal mitochondrial and endoplasmic reticulum (ER), and activated inflammatory pathways (Sapp et al. 2014). Further differences between fructose and glucose metabolism were found at the molecular level. For instance, a reduction in the activity of the hepatic mitochondrial β -oxidation by fructose, but not glucose, was correlated with a decrease in the activity of peroxisome proliferator-activated receptor- α (PPAR α), a nuclear receptor that plays a pivotal role in fatty catabolism (Rebollo et al. 2014), oxidative stress, and inflammatory response (Nan et al. 2014). Effects of fructose and glucose on cell cycle regulation in whole liver and skeletal muscle of rats have been reported using mini-array analysis (Fernández-Novell et al. 2014). Rats in this study received either 250 mg glucose/100 g body weight or 280 mg fructose/100 g body weight and the phosphorylation of several proteins was assessed. In the liver, glucose load induced a very intense increase in the serine phosphorylation levels of cyclin D3, total protein kinase C (PKC), ERK-2, protein phosphatase protein tyrosine phosphatase 1B (PTP1B), c-kit, and clathrin. The fructose load induced a significant increase in serine phosphorylation ERK-2, PTP2, cyclin D3, PI3 kinase/p85, and clusterin (Fernández-Novell et al. 2014).

Although many studies report differential activities of dietary fructose and glucose, some have found that fructose and glucose do have similar effects. Johnston and colleagues found no difference in liver triacylglycerol or biochemistry in healthy overweight men who ingested 25% fructose or 25% glucose energy diet for 2 weeks, followed by a 6-week “washout period”, and resumed the diets again for 2-week interval (Johnston et al. 2013). Analogous effects of fructose and glucose on lipogenic gene expression and intermediary metabolism were found in human cell line (HepG2) in vitro (Hirahatake et al. 2011).

Global consumption of fructose and glucose

The use of fructose increased from 16% in 1978 to 42% in 1998 in the USA which coincided with an increase of the average daily intake (Vos et al. 2008; Marriott et al. 2009). It has been reported that the mean consumption of fructose has increased and was estimated to be 54.7 g/day and accounted for 10% of total Americans' daily caloric intake (Vos et al. 2008). Additionally, fructose contributes approximately 200–250 kcal/day (sucrose and HFCS are each half fructose), 7–8% of the current 2700 kcal/day *per capita* total calorie intake reported in the USA (White 2008). However, overall estimated carbohydrate intake also increased from 1978 to 2004. This means that the overall estimated mean total fructose intake when expressed as a percentage of carbohydrate intake declined in the same period (Marriott et al. 2009).

Recent reports also indicate that consumption of added sugars, including the HFCS, is decreasing in the USA (since 2005) (White 2008; Welsh et al. 2011). Nevertheless, the highest consumption of fructose (HFCS) is reported in the USA, Western countries (e.g., Canada, the Netherlands, Germany), South Korea, and Japan, while the lowest level of consumption is reported in India (White 2008; Marriott et al. 2009). In Italy, the mean daily intake of total, industrial, and fruit fructose in patients suffering from NAFLD was 18.0 ± 8.7 g, 6.0 ± 4.7 g, and 11.9 ± 7.2 g, respectively (Petta et al. 2013).

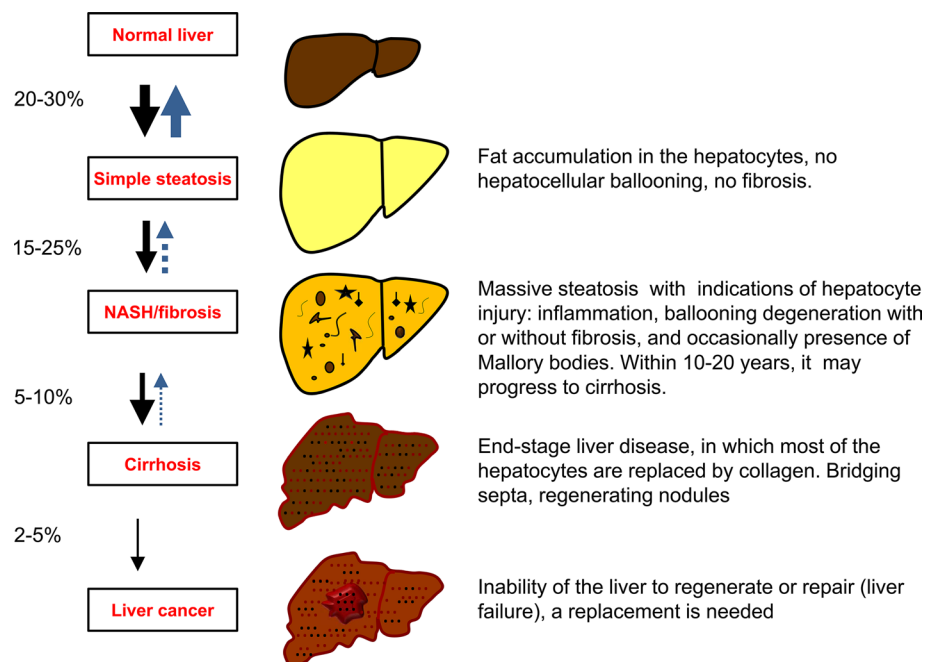
Fructose and NAFLD

Liver metabolism comprises a large variety of interrelated anabolic and catabolic functions which are performed complementarily in different anatomical zones (Gebhardt and Matz-Soja 2014). Simple steatosis is the accumulation of ectopic fat (deposition of triglycerides within cells of non-adipose tissue that normally contain only small amounts of fat) in the hepatocytes (Lettner and Roden 2008). Hepatic steatosis affects 20–30% of Americans and the population of the Western countries, who do not consume excessive alcohol (>30 g/day) (Bellentani et al. 2010). The most common cause of fatty liver can probably be attributed to increased caloric intake, which exceeds the rates of caloric expenditure (Birkenfeld and Shulman 2014). Simple steatosis has long been thought to be a relatively benign state of liver injury; however, clinical relevance indicates that 15–25% of patients could develop NASH. Importantly, up to 20% of patients have long-standing NASH and may develop fibrosis, cirrhosis, or even hepatocellular carcinoma over their lifetime (Ekstedt et al. 2006; Edmison and McCullough 2007; Fig. 1). The reason why a fatty liver progresses to develop inflammation and/or fibrosis, i.e., NASH, which is considered as the most violent form of NAFLD, is still unknown.

Fructose-induced NAFLD in humans

Important risk factors for the development of NAFLD in humans include low-physical activity, dietary factors, gut

Fig. 1 Schematic of progression non-alcoholic fatty liver disease (NAFLD) depicts histological spectrum and estimated prevalence of the disease stages



microbiota, genetic and regulatory factors, and oxidative stress (Matz-Soja et al. 2016; Tilg et al. 2016). Here, we focus on aspects linked to dietary fructose. In general, high energy intake, inadequate energy compensation, and the special metabolism of fructose could explain the association between fructose and NAFLD (Assy et al. 2008; Ouyang et al. 2008; Rebollo et al. 2014).

One-year period of overeating of fruits rich in fructose, increased body weight, blood triglyceride, and transaminase levels in a 33-year-old male patient with insulinoma who was diagnosed NAFLD (Rokutan et al. 2015). Increased fructose intake is also associated with decreased hepatic insulin sensitivity and increased fibrosis severity in patients with NASH (Yki-Järvinen 2010). Similar effects have been seen in human studies where short-term (9d consecutively) fructose intake (25% of energy content) was associated with increased hepatic fatty acid synthesis and liver fat in healthy men fed weight-maintaining diets (Schwarz et al. 2015). In NAFLD patients, increased daily fructose ingestion was strongly associated with low high-density lipoprotein (HDL) cholesterol, increased caloric intake, hyperuricemia, increased fibrosis severity, and increased hepatic inflammation and hepatocyte ballooning (Abdelmalek et al. 2010). Overweight subjects who received fructose-sweetened beverages for 10 weeks had a significant increase in visceral adipose tissue, dyslipidemia, and an impaired glucose tolerance compared to the corresponding glucose cohorts, although weight gain was not different in either cohorts (Stanhope et al. 2009; Tappy et al. 2013). Other work has found that consumption of honey, sucrose, and HFCS (55%) produces similar metabolic effects in glucose-tolerant/glucose-intolerant individuals who consumed 50 g carbohydrate daily from assigned sweeteners (Raatz et al. 2015; Basaranoglu et al. 2015). Despite similar overall energy, the total fructose intake was found to be significantly higher in NAFLD patients compared to healthy individuals (Thuy et al. 2008) further highlighting the significant consequences of fructose consumption in NAFLD. In support of such findings a pilot intervention study focusing on the reduction of fructose intake over 6 months provided some evidence for a significant impact on reduction of intrahepatic fat content of NAFLD patients (Volynets et al. 2013).

Concerning the comparison of fructose and glucose effects the observation that plasma adropin concentrations vary in response to dietary sugar intake in human subjects, irrespective of duration, sex or age, is intriguing. Adropin is a peptide hormone encoded by the *Energy Homeostasis Associated (ENHO)* gene that regulates whether the body burns fat or sugar during feeding and fasting cycles (Butler et al. 2015). Glucose consumption reduced plasma adropin levels, whereas fructose

consumption increased plasma adropin levels. However, the consumption of HFCS as 25% of daily energy requirements had no effect through 10 weeks of the study (Butler et al. 2015). Overall, these studies indicate that both the quality and quantity of ingested carbohydrate are important risk factors for NAFLD and its correlated abnormalities.

Although these findings support the pathological role of fructose, data from human studies about the long-term effects of fructose intake are criticized to have limitations concerning their sample size and retrospective nature (Sloboda et al. 2014) and a recent meta-analysis of six observational and 21 intervention studies carried out by Chung and co-workers concluded that the apparent association between indexes of human health (i.e., liver fat, hepatic de novo lipogenesis and others) and fructose or sucrose intake appear to be confounded by excessive energy intake (Chung et al. 2014). Other meta-analysis studies cited in (Bray and Popkin 2014), however, gave opposite conclusions. Thus, even at the level of meta-analyses, there is no general agreement and large and comprehensive interventions studies are urgently needed to clarify the exact role of fructose in NAFLD and MetS in human subjects.

Animal studies

Animal studies provided additional evidence on the role of fructose-/HFCS-induced hepatic steatosis. In rodents, consumption of fructose has been associated with increased periportal fibrosis, dyslipidemia, ER stress, and impaired insulin signaling (Kawasaki et al. 2009; Ren et al. 2012). Typical features of human NASH including excessive steatosis, hepatocellular ballooning, inflammation, and fibrosis were reproduced by the addition of fructose to saturated fat (Fig. 2) and cholesterol diets in animals (Charlton et al. 2011). Maternal fructose intake resulted in age- and sex-specific changes in fetal, placental, and neonatal free fatty acid metabolism, which may be associated with disruptions in core clock gene machinery in rats (Clayton et al. 2015). Results from many other experimental studies are presented further down in their specific context.

Pathogenic mechanisms

An imbalance in fatty acid synthesis, β -oxidation, and TG exportation processes leads to the accumulation of fat in hepatocytes. High-caloric diets could also contribute in this process. As mentioned, fructose is efficiently converted into glyceraldehyde-3-phosphate by avoiding the rate-controlling step of glucose metabolism in hepatocytes. Thus, the consumption of fructose-enriched diets increased

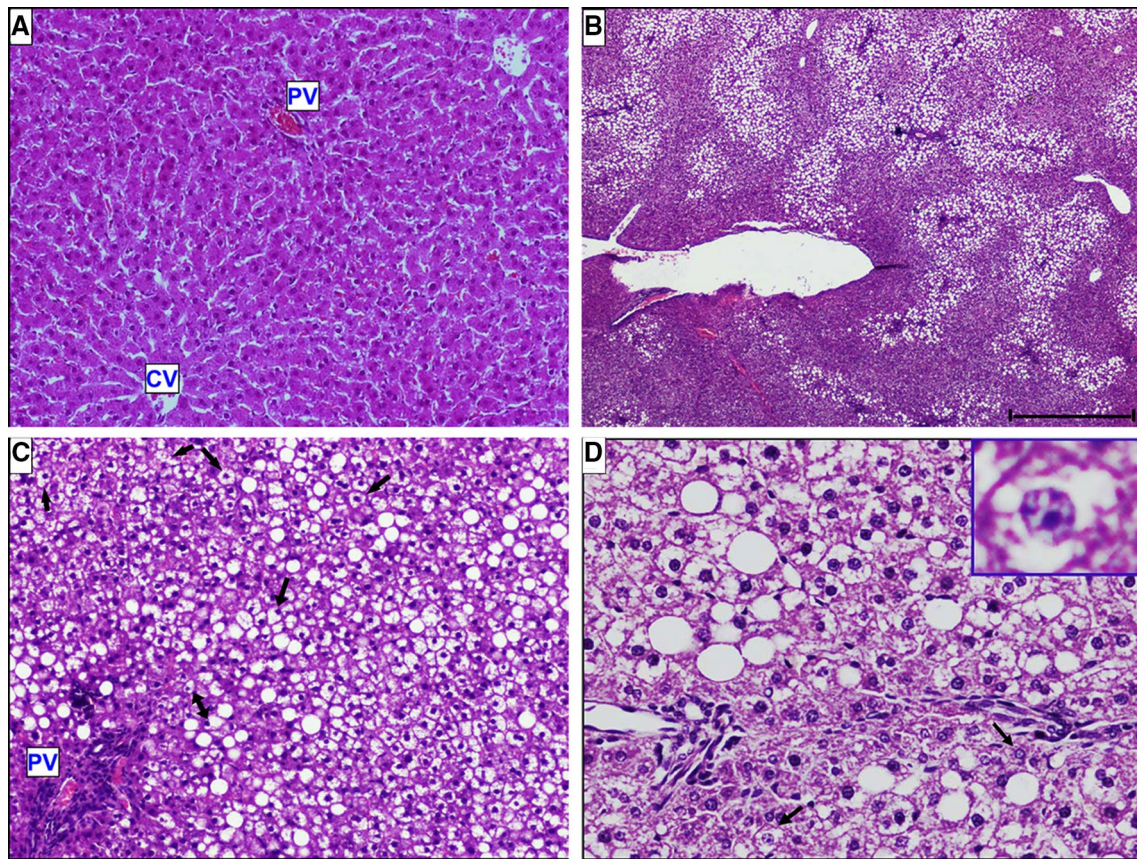


Fig. 2 Representative photomicrographs of HE stained liver sections. **a** “Normal” rat liver structure, **b** an overview microsection of the high-fructose-induced NAFLD ($\times 20$), **c** macrosteatotic hepatocytes and portal inflammation in Zone I, while hepatocytes ballooning with central nuclei in the centrilobular area ($\times 100$), **d** nuclear vacuola-

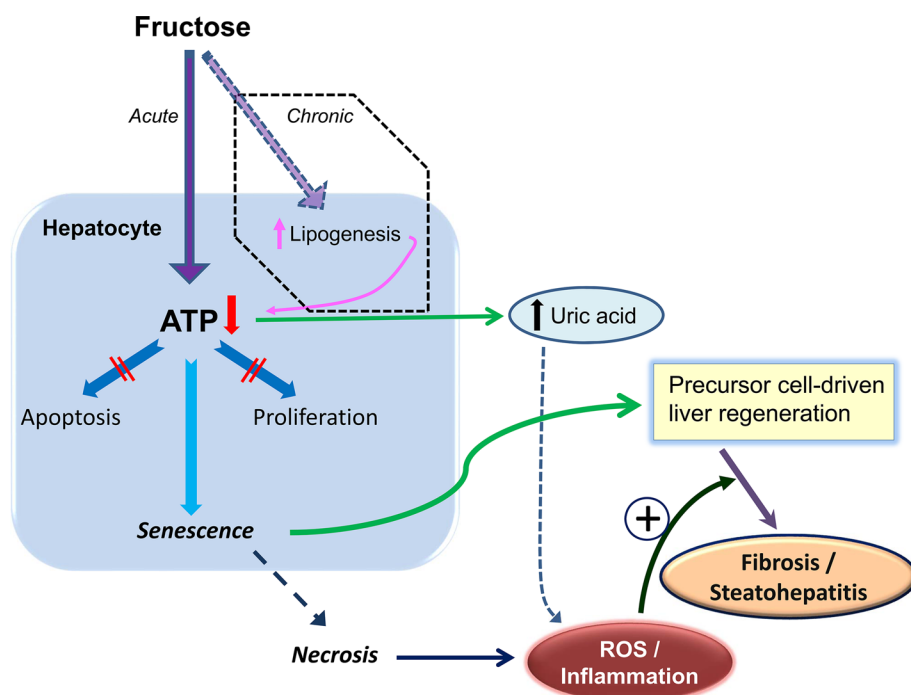
tion of hepatocytes *inset* $\times 200$ and *arrows*. Many hepatocytes show nuclear vacuolation, a feature much more commonly seen when fatty liver has a non-alcoholic etiology. Hepatic simple steatosis in which $>5\%$ of the parenchyma histologically manifest intracytoplasmic TG

de novo lipogenesis and the activities of hepatic lipogenic enzymes fatty acid synthase and stearoyl-CoA-desaturase-1 (Crescenzo et al. 2013). These conditions sensitize the liver to inflammation; a key pathophysiologic feature of NASH (Renaud et al. 2014), and increase mitochondrial coupling leading to oxidative stress (Crescenzo et al. 2013). Another important point in this regard is that high-fat and high-sugar (Western) diet-induced steatohepatitis that was dependent on fructokinase, suggesting a key role for fructose (derived from sucrose) in this development of NASH (Ishimoto et al. 2013). It is, however, not only dietary fructose which causes a metabolic imbalance, but also the endogenous fructose production and metabolism in the liver that was found to contribute to the development of the MetS (Lanaspa et al. 2013). Song et al. stated a novel mechanism for obesity-related fatty liver, in which fructose absorption from the duodenum impaired copper absorption and increased *de novo* lipogenesis (Song et al. 2012).

Fructose-dependent derangement of liver regeneration and ATP depletion

It has been intensively studied that fructose uptake exerts an acute ATP depletion by flooding the hepatocytes with fructose-1-phosphate via fructokinase (Boesch et al. 1997; Latta et al. 2007; Kanuri et al. 2011a). Patients with overt type 2 diabetes mellitus (T2D) can exhibit reduction in hepatic ATP synthesis and impaired repletion of their hepatic ATP stores upon ATP depletion by fructose (Koliaki 2013). Consequently, hepatocytes are prevented from proliferation as well as from entering apoptosis (Speicher et al. 2012). Instead, they may switch to a special senescent phenotype exhibiting abnormal metabolic features (e.g., increased production of urate, steatotic rewiring) and other aberrant cellular functions. Altogether, these changes impair normal liver regeneration and favor the activation of precursor cell-driven regenerative pathways. Since senescent hepatocytes may have a higher risk of dying

Fig. 3 Hypothesis on fructose-dependent derangement of liver regeneration. Acute fructose uptake results in a dramatic drop of hepatocellular ATP levels. This may enhance hepatocytes to become senescent. Altogether these changes impair normal liver regeneration and favor the activation of precursor cell-driven regenerative pathways. ATP depletion results in uric acid over-production, which in turn promotes the formation of reactive oxygen species (ROS) and inflammation. Senescent hepatocytes are more likely to undergo necrotic pathway. Inflammatory reactions and fibrogenesis can be also induced



via necrosis (Rashid et al. 1999), inflammatory reactions may be induced which impact on hepatic precursor cells at all regenerative stages and support fibrotic degeneration as well as progression to steatohepatitis (Fig. 3). Under chronic fructose intake, on the other hand, the main effect, on the hepatocytes, seems to be an insufficient replenishment of ATP (Cortez-Pinto et al. 1999; Abdelmalek et al. 2012). This effect is definitively associated with steatosis, but it is not fully explained yet at molecular levels. Transient depletion of cellular ATP levels were found in cultured primary human hepatocytes (Weiland et al. 2012), but not in human liver cancer cell lines such as HepG2, HuH7, Hep3B, and PLC/PRF/5 (Weiland et al. 2012).

Fructose intake increases the production of uric acid

Using MR imaging and the association between body mass index (BMI) and ATP homeostasis was assessed in healthy subjects with a broad range of BMIs and without any known liver disease (age 18–60 years, and daily alcohol drink ≤ 20 g) (Nair et al. 2003). Reduced hepatic ATP stores were shown to be more prevalent in overweight and obese subjects than in lean subjects (Nair et al. 2003). This can be assessed by the rate of production of uric acid by hepatocytes as it is degraded as a sensitive index of compromised cell ATP homeostasis (Petrie et al. 2013). Fructose differs from other sugars also in its ability to cause intracellular hepatic ATP depletion, nucleotide turnover, and the generation of AMP and uric acid (Lanaspa et al. 2012a; Johnson et al. 2013). Higher dietary fructose was found to associate

with impaired hepatic ATP homeostasis in obese individuals with T2D (Abdelmalek et al. 2012). In this regard, subjects with high uric acid show a greater level of impaired hepatocellular energy homeostasis in response to fructose suggesting that fructose is a risk factor for progressive liver injury (Abdelmalek et al. 2012). In keeping with this, recent studies have shown that consumption of 25% of energy requirements of fructose-sweetened beverages for 10 weeks increases circulating concentrations of uric acid and retinol-binding protein-4 (RBP-4) in overweight/obese humans (Cox et al. 2012). This could lead to further fat deposition as fructose-induced uric acid generation causes mitochondrial oxidative stress that stimulates fat accumulation independent of excessive caloric intake, and may have a causal role in diabetes and obesity (Lanaspa et al. 2012b; Speicher et al. 2012; Johnson et al. 2013). These studies suggest that raised plasma uric acid could be a marker of compromised hepatic ATP homeostasis.

Fructose intake and oxidative stress

Pathological oxidative stress arises due to an imbalance of reactive oxygen species (ROS) and antioxidant production, altering the normal redox state of cells (Barker et al. 2014). ROS sources within the cells include the mitochondria, peroxisomes, NADPH oxidase in the plasma membrane of activated macrophages, and others (Fig. 4). Factors promoting lipid peroxidation and oxidative stress include the induction of CYP2E1 and mitochondrial dysfunction favoring the formation of ROS (Pessayre and Fromenty 2005),

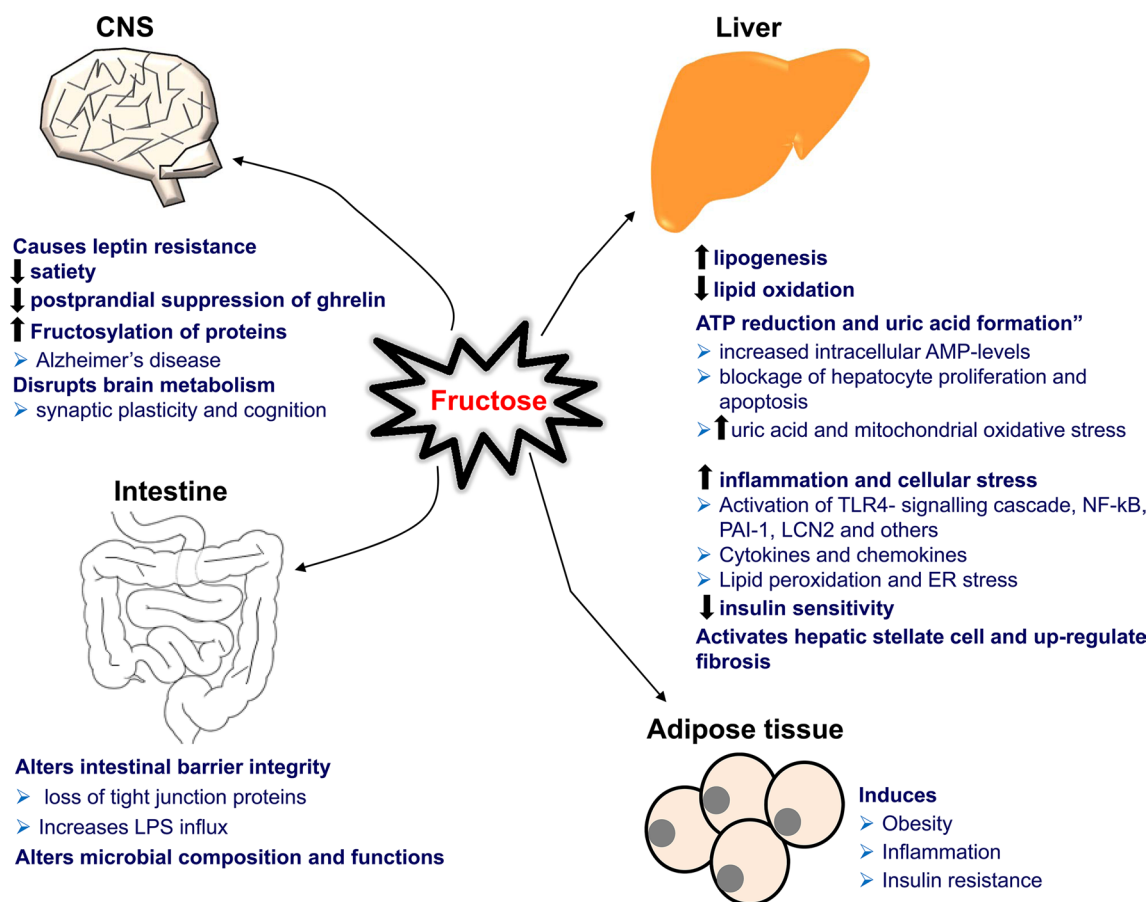


Fig. 4 Potential main deleterious effects of fructose intake

increased lipid oxidation by-products and the cytokines. Both oxidative stress and ER stress, which are important factors in the progression of simple steatosis to NASH, were able to independently cause hepatic lipid accumulation. Treatment of larvae of zebra fish with fructose-containing culture medium which also includes valinomycin (an oxidative stress-inducing agent), or tunicamycin, (an ER stress-inducing agent) resulted in the accumulation of lipid in the liver. The fructose-treated fish exhibited similar expression profiles of lipogenic and inflammatory genes seen in the valinomycin-treated group (Sapp et al. 2014). Fructose also induces an imbalance in gut microbiota and bacterial by-products that can promote the onset of NAFLD and its correlated abnormalities (see below). Exendin-4 (GLP-1 analogs) reduces human primary hepatocyte steatosis and improves survival by enhancing the unfolded protein response and promoting macroautophagy by decreasing the oxidative stress (Sharma et al. 2011). Lim et al. describes that NAFLD is characterized by two steps of liver injury (the ‘two-hit’ theory). In the first ‘hit,’ hepatic metabolism of fructose promotes intrahepatic lipid accumulation (steatosis), inhibition of mitochondrial

β -oxidation of long-chain fatty acids, hepatic and skeletal muscle insulin resistance, and hyperglycemia. In the second ‘hit,’ owing to the molecular instability of its five-membered furanose ring, fructose promotes protein fructosylation and formation of ROS, which require quenching by hepatic antioxidants (Lim et al. 2010).

Insulin resistance

IR is known to be an underlying cause of T2D, cardiovascular disease, and the MetS (Kelishadi et al. 2014). Physiologically, IR is a condition where higher than normal insulin levels are needed to adequately transport glucose due to the decreased response of peripheral tissues, and to suppress the hepatic glucose production. As the liver is a major depot for glucose uptake and storage, hepatic IR leads to further elevation in blood glucose levels (Michael et al. 2000). Fructose ingestion reduces insulin sensitivity. In young rats given fructose-enriched drinks for 12 weeks, there was a significant increase in fasting glycemia, systolic blood pressure, and presented IR (higher HOMA-IR) compared to controls (Dupas et al. 2016).

Hepatic IR is also a major contributor to steatosis in the pathogenesis of NAFLD, which could be attributed to impaired insulin-stimulated IRS-1/2 tyrosine phosphorylation, and the expression of the microsomal CYP2E1 is important in this context (Samuel et al. 2004). As a consequence of excessive fructose-triggered lipid accumulation in the hepatocytes; harmful conditions such as mitochondrial dysfunction, ER stress, inflammation, and hepatic IR could be occurred (Birkenfeld and Shulman 2014; Pyo and Lee 2014).

Fructose and gut microbiome and inflammation

Emerging evidences demonstrate that intestinal microbiome is among the key components in the progression of NAFLD. Due to its anatomical links to the gut, the liver is constantly exposed to gut-derived endotoxins delivered via portal vein and thus functions as the bodies first line of defense (Mouzaki et al. 2013). Kupffer cells are the resident macrophages in the sinusoids and are able to process bacteria and endotoxins, thereby playing a major role in the protection against septicemia. More than 90% of bacterial phylogenetic types encompassing the mouse distal gut microbiota belong to the Bacteroidetes and Firmicutes phyla, similar to that in human beings (Frank et al. 2007). Aron-Wisniewsky et al. reported that intestinal-bacterial patterns appear to vary between obese and lean humans (Aron-Wisniewsky et al. 2013), and patients with NAFLD have more frequently bacterial overgrowth than controls (Wigg et al. 2001).

Mechanism of fructose-induced bacterial translocation and the onset of NAFLD

The consumption of fructose may contribute to the onset of NAFLD through mechanisms involving an increased translocation of intestinal-microbial components, altered concentration of TLR, and activation of TLR-dependent signaling pathways (Spruss et al. 2009). Indeed, feeding monkeys with fructose for 6 weeks caused a rapid liver damage secondarily to microbial translocation and increased incidence of T2D (Kavanagh et al. 2013). In fructose-fed *Tlr4*^{-/-} mice; hepatic steatosis, lipid peroxidation, and TNF α levels were significantly reduced compared to fructose-fed wild-type mice (Spruss et al. 2009). A long-term intake of fructose was associated with a marked reduction in the protein concentration of the tight junction protein in the duodenum (Spruss et al. 2012b). Mice fed HFD and fructose together showed a higher infiltration of lymphocytes into the liver and a lower inflammatory profile of Kupffer cells than mice fed HFD alone. The dysbiosis (a microbial imbalance) associated with dieting showed

that fructose specifically prevented the decrease in intestinal bacteria in HFD-fed mice and increased *Erysipelotrichi* in mice fed with fructose, independently of the amount of fat. These data highlight that the complexity of diet composition could impact the development of liver lesions during obesity (Dupas et al. 2016). The number of members of two bacterial genera, *Coproccoccus* and *Ruminococcus*, were increased by the fructose-rich diet and reduced by both antibiotic and fecal treatments in rats, pointing to a correlation between their abundance and the development of the MetS (Di Luccia et al. 2015). These studies therefore indicate the importance of gut microbiota in fructose-induced NAFLD and T2D.

Postprandial acute effects of fructose consumption were investigated as well. The sucrose- or fructose-fed mice had a low-grade inflammatory response which includes leukocytosis and high serum concentrations of pentraxin-3, leptin, and resistin. An intensified inflammatory response concomitant with an increased liver influx of neutrophils and dropping of the adiponectin concentration was found in mice following a fructose-containing meal (Rodrigues et al. 2014).

Fructose and the brain

The hypothalamus is a major area in the brain where nutritionally relevant information originating from the peripheral organs is integrated (Li et al. 2015).

Fructose effects and potential mechanisms

Fructose feeding up-regulates rat brain GLUT5, suggesting fructose entry. It was reported that fructose-conditioned preferences in rats were based primarily on the sugar's palatable taste, while glucose preferences were based on its taste as well as its postingestive actions. Dopamine antagonists differentially blocked fructose-conditioned preferences acquisition and expression in different regions of the brain (Sclafani and Ackroff 1994; Malkusz et al. 2015).

Endocannabinoids have been implicated in the regulation of the consumption of palatable food. It was found that fructose drinking affects enzymes involved in the metabolism of hypothalamic endocannabinoids and this predates obesity in rats (Erlanson-Albertsson and Lindqvist 2010). The short-term effect of sucrose, glucose, or fructose solutions on peripheral and central appetite signals in rats was also studied. Consuming sugar solutions resulted in increased serum leptin, decreased peptide-YY, and down-regulated hypothalamic *Neuropeptide-Y* mRNA. Fructose drinking increased serum ghrelin. Moreover, consumption of sucrose or fructose solution resulted in up-regulated hypothalamic cannabinoids mRNA (Lindqvist et al. 2008).

Melanocortin receptor-4 (MC4R), a receptor for the α -MSH and leptin in the brain and liver, plays a key role in the regulation of lipid metabolism (Xu et al. 2013). *Mc4r* deficiency increases appetite and diminishes satiety and is the most common cause of monogenic obesity (Albuquerque et al. 2014). In addition, sex differences in brain *Mc4r* expression was found in mice (Qu et al. 2014). Maternal and post-weaning high-fat diet predisposes the offspring for obesity, glucose intolerance, and IR in later life. The hypothalamus expression of *Mc4r* was significantly increased in mouse offspring fed a high-fat, high-sucrose diet during gestation into 32 weeks (Zheng et al. 2015). High-fructose intake during pregnancy modulated the hepatic and hypothalamic AMPK signaling pathways in female offspring after birth (Mukai et al. 2014). Maternal high-fructose diet has also long-term effects on brain mitochondrial function in aging rats, which appears to be linked to an increase in levels of mitochondrial uncoupling protein-5 (Mortensen et al. 2014). Moreover, high-fructose consumption exacerbated the pathology of brain trauma by further disrupting energy metabolism and brain plasticity (Agrawal et al. 2015).

Monosaccharides activate distinct neuronal circuits to promote feeding behavior (Rorabaugh et al. 2014). Compared to glucose, fructose-given *Drosophila* flies were obese, consumed more food, and the brain mRNA expression of insulin-related peptides was reduced (Rovenko et al. 2015). Overnight fasted, normal-weight healthy volunteers without diabetes and with a mean age of 31 years have been given 75 g fructose or glucose during MRI scanning. Consumption of fructose resulted in a distinct pattern of hypothalamic regional cerebral blood flow (CBF) and a smaller increase in systemic glucose, insulin, and glucagon-like polypeptide 1 levels compared with subjects fed an isocaloric amount of glucose (Page et al. 2013). The increased food intake was attributed to the rapid initial steps of fructose metabolism in the brain, provoking an immediate drop in the ATP/AMP ratio, increased AMPK activity, decreased acetyl-CoA-carboxylase activity, and malonyl-CoA amount in the hypothalamus (Cha et al. 2008).

Substances that could counteract fructose-induced brain deterioration

Lycopene, a member of the carotenoid family of phytochemicals, exhibited protective effects against brain damage resulting from fructose-caused IR, neuro-inflammation, and reduced activity of cholinergic system in rats (Yin et al. 2014). Nutrient betaine (a metabolite of choline and a component of methionine–homocysteine cycle) has protective effects against diet-induced NASH and neural injury. Interestingly, betaine inhibited fructose-caused hypothalamic astrogliosis by inhibiting TLR4/NF- κ B pathway

and histone deacetylases-3 overexpression in rats (Li et al. 2015). Analogously, increasing angiotensin-(1–7) levels in the brain were found to attenuate cardiovascular and metabolic disorders in fructose-fed rats (Guimaraes et al. 2014). Moreover, curcumin (an effective scavenger of ROS and nitrogen species) protected against neuronal damage in hippocampal dentate gyrus of fructose-fed mice by inhibiting microglia activation and suppressed fractalkine/CX3C-Receptor1 up-regulation in the neuronal network (Xu et al. 2016).

Fructose and alcohol

The hepatocytes play important functions in detoxification of poisons, e.g., alcohol (Pastoret et al. 2013). As introduced, NAFLD encounters none or mild-to-moderate alcohol consumption. The cutoff value for a tolerable alcohol consumption in NAFLD suspected patients ranges from absence, >1 drink/day (10 g alcohol) to 30 g/day (Dam-Larsen et al. 2004).

Similarities in metabolism and post-ingestion consequences

Lustig (Lustig 2010) has reported that there are similarities in fructose and ethanol metabolism in the liver in these aspects:

- Both serve as substrates for *de novo* lipogenesis and promote hepatic IR, dyslipidemia, and hepatic steatosis
- Fructosylation of proteins with resultant superoxide formation similar to acetaldehyde, an intermediary metabolite of ethanol
- Promoting sensations of hunger via blocking of leptin signaling.

Nevertheless, a few studies have described the consequences of combined consumption of fructose and ethanol on liver pathology or their potential synergistic effects.

Acute concurrent consumption

The acute effect of fructose administered intravenously or orally with an alcoholic drink on liver metabolism or alcohol clearance from bloodstream has been investigated clinically (Boesch et al. 1997; Onyesom 2005). Fructose had beneficial effects against alcoholic intoxication, whereby it caused a significant increase in the rate of decrease of blood ethanol levels (Brown et al. 1972; Uzuegbu and Onyesom 2009). It has been reported that honey (naturally contains fructose) reduced blood alcohol concentration but did not affect the level of serum malondialdehyde (MDA, a marker for lipid peroxidation) and glutathione peroxidase

activity in intoxicated male mice (Shi et al. 2015). In mice, pharmacological activation of aldehyde dehydrogenase-2 by Alda-1 reversed liver damage through accelerating aldehyde clearance, and upregulated fatty acid catabolic enzymes and thus reversed steatosis (Zhong et al. 2015). Interestingly, it was suggested that fructose stimulates ethanol oxidation, increasing the rate of oxidative phosphorylation. Consequently, the capacity of the respiratory chain for oxidizing reducing equivalents derived from ethanol is enhanced (Scholz and Nohl 1976). The change in alcohol metabolism in the presence of fructose has been linked to the ratio of NAD^+/NADH , which facilitates alcohol oxidation by a complementary mechanism (Alwahsh et al. 2014a).

Chronic ethanol and fructose consumption

Long-term ingestion of fructose and ethanol together exaggerated liver dysfunction (Song et al. 2016), dyslipidemia, and IR compared to each diet separately (Alwahsh et al. 2014a). In NAFLD patients, fasting blood ethanol was found to be at a high level, which is suggested to be due to an increased endogenous ethanol synthesis in the gastrointestinal tract. This increase in blood ethanol levels in NAFLD patients may result from insulin-dependent impairments of alcohol dehydrogenase (ADH) activity in liver tissue rather than from an increased endogenous ethanol synthesis (Engstler et al. 2015). Studies on mouse primary hepatocytes have shown individual exposure to ethanol or fructose did not sensitize the hepatocytes to $\text{TNF}\alpha$ -induced necroptosis. However, concurrent exposure of hepatocytes to fructose and ethanol synergistically promoted mitochondrial sensitivity to $\text{TNF}\alpha$ (Shulga and Pastorino 2014). It has also been demonstrated that CYP2E1 plays an important role in ethanol metabolism, and CYP2E1 inhibition by chlormethiazole blunted ethanol-induced lipid peroxidation, oxidative stress, and the inflammatory response in human hepatoma cell line expressing CYP2E1 (Mahli et al. 2015). Interestingly, dioscin, a natural steroid saponin, widely exists in many herbs and has demonstrated protective effects against ethanol-induced liver injury (Xu et al. 2014). Plasma ceruloplasmin activity and plasma copper levels were shown to be significantly lower in the mice fed with either high-fructose-diet or alcohol (Song et al. 2016). In contrast, liver copper levels as well as plasma copper levels were significantly higher in mice fed an alcohol-plus-high-fructose-containing diet compared to mice fed chronic alcohol alone (Song et al. 2016). Studying fructose and ethanol metabolism will increase our understanding of NAFLD and ALD pathogenesis and their clinical overlap.

Mediators that are altered in fructose-onset NAFLD

Here we underscored selective markers of which their protein expression and levels are changed in plasma and/or in liver (Table 1).

Lipocalin-2 (LCN2) is a 25-kDa secretory glycoprotein originally detected in neutrophil granulocytes and tissues that are exposed to microorganisms (Ahmad et al. 2014). LCN2 plays roles in the acute-phase response, apoptosis, and tissue homeostasis (Borkham-Kamphorst et al. 2011). LCN2 has been also described as a critical regulator of energy metabolism (Guo et al. 2010). Additionally, LCN2 has been proposed as a biomarker to discriminate between sample steatosis and fructose-induced NASH in rats (Alwahsh et al. 2014b).

Plasminogen activator inhibitor-1 (PAI-1) is also an acute-phase protein, the expression of which was increased in the liver and plasma, and correlated with endotoxin concentrations and total carbohydrates intakes in NAFLD patient compared to controls (Thuy et al. 2008). Experimentally, it has been shown that *Pai-1*^{-/-} mice are protected from weight gain and marked hepatic lipid accumulation in a model of high-fat/high-carbohydrate feeding (Kanuri et al. 2011b). PAI-1 has a causal role in mediating the early phase of fructose-induced liver damage through signaling cascades downstream of Kupffer cells and tumor

Table 1 Biomarkers known to change during fructose-induced fatty liver disease

Parameter	Presence	
ALT	Serum	Kanuri et al. (2011b)
Leptin	Serum	Alwahsh et al. (2014b)
Insulin	Serum	Alwahsh et al. (2014a)
RBP-4	Serum	Cox et al. (2012)
Uric acid	Serum and liver	Abdelmalek et al. (2012), Johnson et al. (2013)
Adiponectin	Serum	Spruss et al. (2012a)
MDA	Serum	Volynets et al. (2010)
Homocysteine	Serum	Li et al. (2013)
PAI-1	Serum and liver	Kanuri et al. (2011b)
LCN2	Serum and liver	Alwahsh et al. (2014b)
$\text{TNF-}\alpha$	Serum and liver	Kanuri et al. (2011a)
4-HNE	Liver	Volynets et al. (2010)
TLR4	Liver	Spruss et al. (2009)
iNOS	Liver	Spruss et al. (2011)
Sirt1	Liver	Feige et al. (2008)

RBP-4 retinol-binding protein-4, *MDA* malondialdehyde, *4-HNE* 4-hydroxynonenal, *iNOS* nitric oxide synthase

necrosis factor- α (TNF- α) (Kanuri et al. 2011b). Similar to *nitric oxide synthase* (iNOS), PAI-1 may be a critical factor in mediating the effect of the increased translocation of bacterial endotoxin and the subsequent activation of toll-like receptor (TLR)-4-dependent signaling cascades found in livers of mice chronically exposed to fructose (Kanuri et al. 2011b). The sex-specific differences in the susceptibility to the development of NAFLD were associated with the regulation of the adiponectin-AMPK-PAI-1 signaling, in which the expression of hepatic PAI-1 was markedly induced in female mice (Spruss et al. 2012a).

In a related context, fructose-induced hepatic steatosis and increased translocation of endotoxins were associated with an increased formation of ROS in the liver and expression of hepatic TNF- α , a pro-inflammatory cytokine, secreted mainly by activated macrophages (Bergheim et al. 2008). TNF- α both promotes and is activated by IR, and it also antagonizes adiponectin, an anti-inflammatory adipocytokine (Tiniakos et al. 2010). Furthermore, TNF- α receptor 1^{-/-} (*Tnfr1*^{-/-}) mice were protected from the onset of fructose-induced steatosis and IR (Kanuri et al. 2011a), highlighting the essential role of TNF- α in triggering of NAFLD (Tomita et al. 2006; Martius et al. 2014). Conversely, it was reported that fructose protected murine hepatocytes from TNF-induced mitochondrial apoptosis by modulating JNK signaling, thereby blocking bid-mediated activation in a protein kinase A-dependent manner (Speicher et al. 2012).

Leptin (Ob) is a 16-kDa protein primarily secreted by adipocytes. Leptin affects the immune system and plays a remarkable role in the regulation of lipid and glucose metabolism in the liver (Lou et al. 2010) through the leptin receptor (Ob-R) (Lou et al. 2010). High-fructose-enriched diet had significantly elevated serum leptin levels in fasted rats (Alwahsh et al. 2014a, b). An orally given leptin potentiated a fructose-induced (1) increase in blood glucose and mRNA levels of key gluconeogenesis enzymes, and (2) an increase in blood triglycerides and hepatic lipogenic gene expression. Dietary fructose triggers the release of gastric leptin which, in turn, up-regulates GLUT5 and concurrently modulates metabolic functions in the liver. This could explain, in part, the rapid lipogenic and deleterious effects of fructose (Sakar et al. 2009).

Increased serum *uric acid* levels serve as a biomarker for increased fructose consumption (Abdelmalek et al. 2012). One additional mediator is *sirtuin-1* (*Sirt1*), a NAD⁺-dependent deacetylase that is modulated by cellular redox, and has protective effects. The activation of hepatic SIRT1 enhanced fatty acid oxidation by reducing PPAR α expression and activity, protecting from diet-induced fatty liver, obesity, and IR (Feige et al. 2008). However, the hepatic

Sirt1 activity was decreased in fructose-fed rats, and in human hepatocytes and hepatoma (HepG2) cells treated 25 mM fructose (Rebollo et al. 2014).

Homocysteine levels were found to be higher in patients with stenotic vessels and coronary artery disease scores, and was in fact highest in diabetic patients (Okada et al. 1999). Rats fed a obesogenic diet (high fat, high fructose) had higher homocysteine levels after 5 weeks as compared to chow-fed rats (Oron-Herman et al. 2003; Li et al. 2013).

Therapeutic trials that target fructose-induced NAFLD

We categorized selective therapeutic agents that were used to attenuate fructose-induced fatty liver disease based on their mechanism(s) (Table 2).

Through reducing portal endotoxin levels

This group includes these substances: metformin (dimethylbiguanide) (Lin et al. 2000; Nair et al. 2004; Spruss et al. 2012b), an AMPK-activating drug used in the treatment of T2D and antibiotics (Bergheim et al. 2008), bovine lactoferrin (Ltf), a 76-kDa iron-transporting glycoprotein which is released by neutrophils and mediates anti-inflammatory effects (binds the LPS) (Rivera et al. 2013), bile acids (Volynets et al. 2010), and WAY-362450, a potent synthetic agonist to farnesoid-X-receptor (a member of the nuclear receptor superfamily).

By reducing the expression of lipogenic and inflammatory genes and the oxidative stress

This group includes rapamycin, an inhibitor of mTOR signaling. The “mechanistic target of rapamycin” (mTOR) is a central regulator of a plethora of processes, e.g., cell proliferation and metabolism (Makki et al. 2014). It also includes troglitazone; a new hypoglycemic drug, Levocetirizine; an antihistamine, Lycopene, Carnosine and α -Tocopherol which are antioxidants, and Pentoxifylline; a phosphodiesterase inhibitor.

Through decreasing fructokinase expression

Atorvastatin, a hydroxymethyl-glutaryl-CoA reductase inhibitor, can be used safely in patients with NAFLD and alleviates hepatic steatosis. Atorvastatin decreased the expression of fructokinase in livers of fructose-supplemented rats, reducing the metabolic burden on the liver that is imposed by continuous fructose ingestion (Vilà et al. 2011).

Table 2 Selective therapeutic trials that target fructose-induced NAFLD

Drug/agent	Experimental design	Curative effects
Metformin (Spruss et al. 2012b; Liu et al. 2014a)	C57BL/6J mice were given 30% fructose $\pm$ 300 mg/kg metformin for 8 weeks	Metformin treatment was associated with a reduction of portal endotoxin level (by serving of tight junction proteins, e.g., occludin in the duodenum), hepatic TG and serum ALT activity
Antibiotics (Bergheim et al. 2008)	CD-1 mice were fed with high-fat +15% fructose diet $\pm$ 500 mg/kg metformin for 10 weeks	Metformin alleviated IR
Lactoferrin (Ltf) (Morishita et al. 2013; Li and Hsieh 2014)	C57BL/6 mice drank sugas-sweetened water (30%) $\pm$ 92 mg/kg polymyxin B and 216 mg/kg neomycin for 8 weeks	Antibiotics protect against fructose-induced fatty liver and endotoxemia
	C57BL/6 mice were fed HFCS $\pm$ Ltf at 50, 100, or 200 mg/kg for 8 weeks	Ltf treatment dampens hepatic fat accumulation and lipid peroxidation, scavenges LPS in circulation and subsequent expression of inflammatory cytokines, and improves IR
Bile acids (Volynets et al. 2010)	ICR mice were administered orally with 200 mg/mL/day Ltf for 4 weeks	Ltf reduces visceral fat and liver triglycerides, and thus it had anti-adipogenic activity
WAY-362450 (Liu et al. 2014b)	C57BL/6J mice treated with chenodeoxycholic and cholic acids (100 mg/kg each) $\pm$ a 30% fructose for 8 weeks	Bile acids prevent fructose-induced hepatic steatosis by protection against the fructose-induced translocation of intestinal-bacterial endotoxin
Rapamycin (Makki et al. 2014; Sapp et al. 2014)	C57BL/6J mice, fed 30% fructose were treated $\pm$ 30 mg/kg WAY for 20 days	WAY reduced serum and hepatic TG levels and attenuated the infiltration of immune cells to the liver and gut LPS translocation
	Zebrafish larvae were treated with 4% fructose/glucose $\pm$ 100 nM rapamycin. A 1 μ M tunicamycin or 100 ng/mL valinomycin was added to the treatment as another trail	Rapamycin reversed NASH and decreased the TorC1 activity and phosphorylation of its downstream target. It also reversed oxidative stress-induced hepatic steatosis, and the elevation of lipogenic and inflammatory genes
Levodocetirizine (Shawky et al. 2014)	C57BL/6JRj mice were reared on high-fat-diet for 5 weeks, then 2 mg/kg rapamycin was <i>i.p.</i> administered weekly for 22 weeks	Rapamycin treatment decreased the body weight, protected against IR, and improved the inflammatory profiles of both adipose tissue and liver
Carnosine + α -tocopherol (Giriş et al. 2014)	Male Sprague–Dawley rats were fed 60% fructose $\pm$ Levocetirizine for 8 weeks	Levodocetirizine ameliorates IR, hepatic steatosis, and lipid peroxidation, and improves glucose tolerance
Pentoxifylline (PTX) (Azhar and El-Bassossy 2014)	Rats fed 60% fructose with CAR (2 g/L) $\pm$ TOC (200 mg/kg, <i>i.m.</i> twice a week) for 8 weeks	CAR + TOC treatment decreased IR, NASH, lipid peroxidation, and pro-oxidant in the liver
PLCP (Liu et al. 2014a)	Rats fed HFD and 10% fructose for 12 weeks. PTX was administered (30 mg/kg/day) during the last 4 weeks of the study	PTX alleviates cardiac ischemia and dysfunction following experimental angina in IR through inhibition of the low-grade inflammation
MFGEs (Pyo and Lee 2014)	CD-1 mice gavaged daily with ethanol extract PLCP for 10 weeks	A diet formula of PLCP alleviates IR, NAFLD, hyperlipidemia, and antioxidant status
	Male Sprague–Dawley rats fed 60% fructose for 16 weeks and orally administered with MFGEs (300 mg/kg/day) or atorvastatin (20 mg/kg) for the last 4 weeks of the study	MFGEs enriched with ubiquinones CoQs exert an anti-diabetic effect in T2D by improving IR and hepatic antioxidant enzymes

Table 2 continued

Drug/agent	Experimental design	Curative effects
(–)Epicatechin (EC) (Bettaieb et al. 2014)	Rats fed a chow diet ± EC and 10% fructose for 8 weeks	EC supplementation inhibited events that contribute to IR: increased activity of NADPH oxidase and redox-sensitive signals, expression of NF-κB-regulated proinflammatory cytokines, and ER stress signaling, thus prevented MetS-associated IR
ZSP (Zhao et al. 2014)	Mice fed 20% fructose water ±400 mg/kg ZSP i.g. for 4 weeks	ZSP has hepatoprotective, anti-diabetic and anti-lipidemic effects
Silymarin (SYM) (Prakash et al. 2014)	Wistar rats fed fructose diet for 12 weeks (SYM were given orally during the last 3 weeks)	SYM has anti-hepatotoxic properties; prevented fructose-enhanced hepatic expression of PGC-1α/β and its target transcription factors. It also counteracted the fructose-mediated endothelial dysfunction, metabolic and inflammatory changes
Atorvastatin (Vilà et al. 2011)	Fructose (10% w/v solution) and fructose + atorvastatin (30 mg/kg/day) Sprague–Dawley rats were killed after 2 weeks	Atorvastatin treatment completely abolished histological signs of necroinflammation, reducing the hepatic expression of metallothionein-1 and NFκB binding. Atorvastatin reduced plasma and liver triglyceride concentrations, and increased the hepatic activity of the fatty acid β-oxidation system
<i>PLCP</i> Puerariae radix, <i>Lycium barbarum</i> , <i>Crataegus pinnatifida</i> , and <i>Polygonati rhizome</i> , <i>MFGEs</i> monascus-fermented grains, <i>ZSP</i> <i>Zizyphus jujube</i> cv. <i>Shaanbeitanzao</i> Polysaccharid		

Using herbal medications

Several herbs have shown significant beneficial effects in rodent models. For example, plant extracts have protected against fructose-induced metabolic abnormalities. Flavan-3-ol(–)-epicatechin (EC), which belongs to flavonoids, is present in large concentration in fruits and vegetables (Bettaieb et al. 2014). Liquid extract of *Monascus*-fermented grains enriched with ubiquinones (Coenzyme Qs), an acidic heteropolysaccharide extract (ZSP), derived from *Zizyphus jujube* *Shaanbeitanzao*, and grape seed extract exhibited anti-diabetic and anti-lipidemic effects (Suwanaphet et al. 2010; Pyo and Lee 2014). Similar pharmacological hepatoprotective effects were found in the active fraction of *Bridelia ferruginea*, the root-bark ethyl acetate (Bakoma et al. 2014), (3,3-dimethylallyl)-halfordinol isolated from leaves of *Aegle marmelos* (Saravanan et al. 2014), *Pleurotus eryngii* (Ren et al. 2014), silymarin extract, and a grape seed Procyanidin extract (Downing et al. 2015).

Nutritional management and conclusions

NAFLD is a disorder mainly associated with obesity (Niklas et al. 2012). In humans, frequent episodes of regular physical activity were significantly lower in the group of NAFLD patients than control subjects (Thuy et al. 2008). Therefore, lifestyle modification is still the first line of the treatment, which shows proven benefits in lowering ALT in NAFLD patients (Thuy et al. 2008). This is analogous to how diet and exercise are the first line of treatment for obesity for both adults and children. Patients with NAFLD should have especial diet recommendation and balanced meals, in order to lose at least 7% of their weight if overweight, reducing caloric intake, mainly at cost of cholesterol and saturated fatty acids (Carvalhana et al. 2012). Weight loss achieved by bariatric surgery decreases liver fat deposits, inflammation and improve insulin sensitivity (Yki-Järvinen 2010; Alwahsh and Ramadori 2015). Fructose over-consumption should be avoided, and soft drinks are discouraged.

In fatty liver patients, improvements in markers of cardiometabolic risk and liver function, as well as body fat were achieved when the patients reduced their intake of fructose from industrialized foods (Volynets et al. 2013). Antioxidant therapy for the treatment of NAFLD, including NASH, has the potential to alleviate oxidative stress and cell death that promote disease progression (Al-Busafi et al. 2012). Many NASH patients have also micronutrient deficiencies and do not have enough antioxidant capacity to prevent synthesis of ROS, resulting in necroinflammation (Lim et al. 2010).

Blocking fructose transporters into the hepatocyte via the GLUT8 hexose transporter may also be a potential mechanism to prevent fructose-induced hepatic steatosis (DeBosch et al. 2014). Impaired hepatic ATP loss attributable to increased dietary fructose consumption underscores the urgent need for increased public awareness of the risks associated with high-fructose consumption (Abdelmalek et al. 2012). Fortunately, metformin and other aforementioned drugs have promising protective effects against fructose-induced fatty liver and endotoxins influx from the intestine.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

- Abdelmalek MF, Suzuki A, Guy C et al (2010) Increased fructose consumption is associated with fibrosis severity in patients with nonalcoholic fatty liver disease. *Hepatology* 51:1961–1971. doi:10.1002/hep.23535
- Abdelmalek MF, Lazo M, Horska A et al (2012) Higher dietary fructose is associated with impaired hepatic adenosine triphosphate homeostasis in obese individuals with type 2 diabetes. *Hepatology* 56:952–960. doi:10.1002/hep.25741
- Agrawal R, Noble E, Vergnes L et al (2015) Dietary fructose aggravates the pathobiology of traumatic brain injury by influencing energy homeostasis and plasticity. *J Cereb Blood Flow Metab* 36:941–953. doi:10.1177/0271678X15606719
- Ahmad S, Sultan S, Naz N et al (2014) Regulation of iron uptake in primary culture rat hepatocytes: the role of acute-phase cytokines. *Shock* 41:337–345. doi:10.1097/SHK.0000000000000107
- Albuquerque D, Estévez MN, Vígora PB et al (2014) Novel variants in the MC4R and LEPR genes among severely obese children from the iberian population. *Ann Hum Genet* 78:195–207. doi:10.1111/ahg.12058
- Al-Busafi SA, Bhat M, Wong P et al (2012) Antioxidant therapy in nonalcoholic steatohepatitis. *Hepat Res Treat* 2012:947575. doi:10.1155/2012/947575
- Alwahsh SM, Ramadori G (2015) How does bariatric surgery improve type II diabetes? The neglected importance of the liver in clearing glucose and insulin from the portal blood. *J Obes Weight Loss Ther* 5:5–8. doi:10.4172/2165-7904.1000280
- Alwahsh SM, Xu M, Schultze FC et al (2014a) Combination of alcohol and fructose exacerbates metabolic imbalance in terms of hepatic damage, dyslipidemia, and insulin resistance in rats. *PLoS ONE* 9:e104220. doi:10.1371/journal.pone.0104220
- Alwahsh SM, Xu M, Seyhan HA et al (2014b) Diet high in fructose leads to an overexpression of lipocalin-2 in rat fatty liver. *World J Gastroenterol* 20:1807–1821. doi:10.3748/wjg.v20.i7.1807
- Aron-Wisniewsky J, Gaborit B, Dutour A, Clement K (2013) Gut microbiota and non-alcoholic fatty liver disease: new insights. *Clin Microbiol Infect* 19:338–348. doi:10.1111/1469-0691.12140
- Aroor AR, Habibi J, Ford DA et al (2015) Dipeptidyl peptidase-4 inhibition ameliorates Western diet-induced hepatic steatosis and insulin resistance through hepatic lipid remodeling and modulation of hepatic mitochondrial function. *Diabetes* 64:1988–2001. doi:10.2337/db14-0804
- Assy N, Nasser G, Kamayse I et al (2008) Soft drink consumption linked with fatty liver in the absence of traditional risk factors. *Can J Gastroenterol* 22:811–816
- Azhar A, El-Bassossy HM (2014) Pentoxifylline alleviates cardiac ischemia and dysfunction following experimental angina in insulin resistance. *PLoS ONE* 9:e98281. doi:10.1371/journal.pone.0098281
- Bakoma B, Berké B, Eklü-Gadegbeku K et al (2014) Effect of *Bridelia ferruginea* Benth (Euphorbiaceae) ethyl acetate and acetone fractions on insulin resistance in fructose drinking mice. *J Ethnopharmacol* 153:896–899. doi:10.1016/j.jep.2014.03.065
- Barker CE, Ali S, O’Boyle G, Kirby JA (2014) Transplantation and inflammation: implications for the modification of chemokine function. *Immunology* 143:138–145. doi:10.1111/imm.12332
- Basaranoglu M, Basaranoglu G, Bugianesi E (2015) Carbohydrate intake and nonalcoholic fatty liver disease: fructose as a weapon of mass destruction. *Hepatobiliary Surg Nutr* 4:109–116. doi:10.3978/j.issn.2304-3881.2014.11.05
- Bellentani S, Scaglioni F, Marino M, Bedogni G (2010) Epidemiology of non-alcoholic fatty liver disease. *Dig Dis* 28:155–161. doi:10.1159/000282080
- Bergheim I, Weber S, Vos M et al (2008) Antibiotics protect against fructose-induced hepatic lipid accumulation in mice: role of endotoxin. *J Hepatol* 48:983–992. doi:10.1016/j.jhep.2008.01.035
- Bettaieb A, Vazquez Prieto MA, Rodriguez Lanzi C et al (2014) (–)-Epicatechin mitigates high-fructose-associated insulin resistance by modulating redox signaling and endoplasmic reticulum stress. *Free Radic Biol Med* 72:247–256. doi:10.1016/j.freeradbiomed.2014.04.011
- Birkenfeld AL, Shulman GI (2014) Nonalcoholic fatty liver disease, hepatic insulin resistance, and type 2 diabetes. *Hepatology* 59:713–723. doi:10.1002/hep.26672
- Boesch C, Elsing C, Wegmüller H et al (1997) Effect of ethanol and fructose on liver metabolism a dynamic ³¹phosphorus magnetic resonance spectroscopy study in normal volunteers. *Magn Reson Imaging* 15:1067–1077
- Borkham-Kamphorst E, Drews F, Weiskirchen R (2011) Induction of lipocalin-2 expression in acute and chronic experimental liver injury moderated by pro-inflammatory cytokines interleukin-1 β through nuclear factor- κ B activation. *Liver Int* 31:656–665. doi:10.1111/j.1478-3231.2011.02495.x
- Bray GA, Popkin BM (2014) Dietary sugar and body weight: have we reached a crisis in the epidemic of obesity and diabetes? Health be damned! Pour on the sugar. *Diabetes Care* 37:950–956. doi:10.2337/dc13-2085
- Bray GA, Nielsen SJ, Popkin BM (2004) Consumption of high-fructose corn syrup in beverages may play a role in the epidemic of obesity. *Am J Clin Nutr* 89:537–543
- Brown SS, Forrest JA, Roscoe P (1972) A controlled trial of fructose in the treatment of acute alcoholic intoxication. *Lancet* 2:898–899
- Butler AA, St-Onge M-P, Siebert EA et al (2015) Differential responses of plasma adropin concentrations to dietary

- glucose or fructose consumption in humans. *Sci Rep* 5:14691. doi:[10.1038/srep14691](https://doi.org/10.1038/srep14691)
- Carvalho S, Machado MV, Cortez-Pinto H (2012) Improving dietary patterns in patients with nonalcoholic fatty liver disease. *Curr Opin Clin Nutr Metab Care* 15:468–473. doi:[10.1097/MCO.0b013e3283566614](https://doi.org/10.1097/MCO.0b013e3283566614)
- Cha SH, Wolfgang M, Tokutake Y et al (2008) Differential effects of central fructose and glucose on hypothalamic malonyl-CoA and food intake. *Proc Natl Acad Sci USA* 105:16871–16875. doi:[10.1073/pnas.0809255105](https://doi.org/10.1073/pnas.0809255105)
- Charlton M, Krishnan A, Viker K et al (2011) Fast food diet mouse: novel small animal model of NASH with ballooning, progressive fibrosis, and high physiological fidelity to the human condition. *Am J Physiol Gastrointest Liver Physiol* 301:G825–G834. doi:[10.1152/ajpgi.00145.2011](https://doi.org/10.1152/ajpgi.00145.2011)
- Chung M, Ma J, Patel K et al (2014) Fructose, high-fructose corn syrup, sucrose, and nonalcoholic fatty liver disease or indexes of liver health: a systematic review and meta-analysis. *Am J Clin Nutr* 100:833–849. doi:[10.3945/ajcn.114.086314](https://doi.org/10.3945/ajcn.114.086314)
- Clayton ZE, Vickers MH, Bernal A et al (2015) Early life exposure to fructose alters maternal, fetal and neonatal hepatic gene expression and leads to sex-dependent changes in lipid metabolism in rat offspring. *PLoS ONE* 10:e0141962. doi:[10.1371/journal.pone.0141962](https://doi.org/10.1371/journal.pone.0141962)
- Conlon BA, Beasley JM, Aebersold K et al (2013) Nutritional management of insulin resistance in nonalcoholic fatty liver disease (NAFLD). *Nutrients* 5:4093–4114. doi:[10.3390/nu5104093](https://doi.org/10.3390/nu5104093)
- Cortez-Pinto H, Chatham J, Chacko V et al (1999) Alterations in liver ATP homeostasis in human nonalcoholic steatohepatitis: a pilot study. *JAMA* 282:1659–1664
- Cox CL, Stanhope KL, Schwarz JM et al (2012) Consumption of fructose—but not glucose-sweetened beverages for 10 weeks increases circulating concentrations of uric acid, retinol binding protein-4, and gamma-glutamyl transferase activity in overweight/obese humans. *Nutr Metab* 9:68. doi:[10.1186/1743-7075-9-68](https://doi.org/10.1186/1743-7075-9-68)
- Crescenzo R, Bianco F, Falcone I et al (2013) Increased hepatic de novo lipogenesis and mitochondrial efficiency in a model of obesity induced by diets rich in fructose. *Eur J Nutr* 52:537–545. doi:[10.1007/s00394-012-0356-y](https://doi.org/10.1007/s00394-012-0356-y)
- Dam-Larsen S, Franzmann M, Andersen IB et al (2004) Long term prognosis of fatty liver: risk of chronic liver disease and death. *Gut* 53:750–755
- DeBosch BJ, Chen Z, Saben JL et al (2014) Glucose transporter 8 (GLUT8) mediates fructose-induced de Novo lipogenesis and macrosteatosis. *J Biol Chem* 289:10989–10998. doi:[10.1074/jbc.M113.527002](https://doi.org/10.1074/jbc.M113.527002)
- Di Luccia B, Crescenzo R, Mazzoli A et al (2015) Rescue of fructose-induced metabolic syndrome by antibiotics or faecal transplantation in a rat model of obesity. *PLoS ONE* 10:e0134893. doi:[10.1371/journal.pone.0134893](https://doi.org/10.1371/journal.pone.0134893)
- Downing LE, Heidker RM, Caiozzi GC et al (2015) A grape seed procyanidin extract ameliorates fructose-induced hypertriglyceridemia in rats via enhanced fecal bile acid and cholesterol excretion and inhibition of hepatic lipogenesis. *PLoS ONE* 10:e0140267. doi:[10.1371/journal.pone.0140267](https://doi.org/10.1371/journal.pone.0140267)
- Dupas J, Goanvec C, Feray A et al (2016) Progressive induction of type 2 diabetes: effects of a reality-like fructose enriched diet in young wistar rats. *PLoS ONE* 11:e0146821. doi:[10.1371/journal.pone.0146821](https://doi.org/10.1371/journal.pone.0146821)
- Edmison J, McCullough AJ (2007) Pathogenesis of non-alcoholic steatohepatitis: human data. *Clin Liver Dis* 11:75–104. doi:[10.1016/j.cld.2007.02.011](https://doi.org/10.1016/j.cld.2007.02.011)
- Ekstedt M, Franzén LE, Mathiesen UL et al (2006) Long-term follow-up of patients with NAFLD and elevated liver enzymes. *Hepatology* 44:865–873. doi:[10.1002/hep.21327](https://doi.org/10.1002/hep.21327)
- Engstler AJ, Aumiller T, Degen C et al (2015) Insulin resistance alters hepatic ethanol metabolism: studies in mice and children with non-alcoholic fatty liver disease. *Gut* 65:1564–1571. doi:[10.1136/gutjnl-2014-308379](https://doi.org/10.1136/gutjnl-2014-308379)
- Erlanson-Albertsson C, Lindqvist A (2010) Fructose affects enzymes involved in the synthesis and degradation of hypothalamic endocannabinoids. *Regul Pept* 161:87–91. doi:[10.1016/j.regpep.2010.01.003](https://doi.org/10.1016/j.regpep.2010.01.003)
- Feige JN, Lagouge M, Canto C et al (2008) Specific SIRT1 activation mimics low energy levels and protects against diet-induced metabolic disorders by enhancing fat oxidation. *Cell Metab* 8:347–358. doi:[10.1016/j.cmet.2008.08.017](https://doi.org/10.1016/j.cmet.2008.08.017)
- Fernández-Novell JM, Ramió-Lluch L, Orozco A et al (2014) Glucose and fructose have sugar-specific effects in both liver and skeletal muscle in vivo: a role for liver fructokinase. *PLoS ONE* 9:e109726. doi:[10.1371/journal.pone.0109726](https://doi.org/10.1371/journal.pone.0109726)
- Frank DN, St Amand AL, Feldman RA et al (2007) Molecular-phylogenetic characterization of microbial community imbalances in human inflammatory bowel diseases. *Proc Natl Acad Sci USA* 104:13780–13785. doi:[10.1073/pnas.0706625104](https://doi.org/10.1073/pnas.0706625104)
- Gebhardt R, Matz-Soja M (2014) Liver zonation: novel aspects of its regulation and its impact on homeostasis. *World J Gastroenterol* 20:8491–8504. doi:[10.3748/wjg.v20.i26.8491](https://doi.org/10.3748/wjg.v20.i26.8491)
- Giriş M, Doğru-Abbassoğlu S, Kumral A et al (2014) Effect of carnitine alone or combined with α -tocopherol on hepatic steatosis and oxidative stress in fructose-induced insulin-resistant rats. *J Physiol Biochem* 70:385–395. doi:[10.1007/s13105-014-0314-7](https://doi.org/10.1007/s13105-014-0314-7)
- Gorden DL, Ivanova PT, Myers DS et al (2011) Increased diacylglycerols characterize hepatic lipid changes in progression of human nonalcoholic fatty liver disease; comparison to a murine model. *PLoS ONE* 6:e22775. doi:[10.1371/journal.pone.0022775](https://doi.org/10.1371/journal.pone.0022775)
- Guimaraes PS, Oliveira MF, Braga JF et al (2014) Increasing angiotensin-(1–7) levels in the brain attenuates metabolic syndrome-related risks in fructose-fed rats. *Hypertension* 63:1078–1085. doi:[10.1161/HYPERTENSIONAHA.113.01847](https://doi.org/10.1161/HYPERTENSIONAHA.113.01847)
- Guo H, Jin D, Zhang Y et al (2010) Lipocalin-2 deficiency impairs thermogenesis and potentiates diet-induced insulin resistance in mice. *Diabetes* 59:1376–1385. doi:[10.2337/db09-1735](https://doi.org/10.2337/db09-1735)
- Hirahatake KM, Meissen JK, Fiehn O, Adams SH (2011) Comparative effects of fructose and glucose on lipogenic gene expression and intermediary metabolism in HepG2 liver cells. *PLoS ONE* 6:e26583. doi:[10.1371/journal.pone.0026583](https://doi.org/10.1371/journal.pone.0026583)
- Ishimoto T, Lanasa MA, Rivard CJ et al (2013) High-fat and high-sucrose (western) diet induces steatohepatitis that is dependent on fructokinase. *Hepatology* 58:1632–1643. doi:[10.1002/hep.26594](https://doi.org/10.1002/hep.26594)
- Johnson RJ, Nakagawa T, Sanchez-Lozada LG et al (2013) Sugar, uric acid, and the etiology of diabetes and obesity. *Diabetes* 62:3307–3315. doi:[10.2337/db12-1814](https://doi.org/10.2337/db12-1814)
- Johnston RD, Stephenson MC, Crossland H et al (2013) No difference between high-fructose and high-glucose diets on liver triacylglycerol or biochemistry in healthy overweight men. *Gastroenterology* 145:1016–1025.e2. doi:[10.1053/j.gastro.2013.07.012](https://doi.org/10.1053/j.gastro.2013.07.012)
- Kanuri G, Spruss A, Wagnerberger S et al (2011a) Role of tumor necrosis factor α (TNF α) in the onset of fructose-induced non-alcoholic fatty liver disease in mice. *J Nutr Biochem* 22:527–534. doi:[10.1016/j.jnutbio.2010.04.007](https://doi.org/10.1016/j.jnutbio.2010.04.007)
- Kanuri G, Spruss A, Wagnerberger S et al (2011b) Fructose-induced steatosis in mice: role of plasminogen activator inhibitor-1, microsomal triglyceride transfer protein and NKT cells. *Lab Invest* 91:885–895. doi:[10.1038/labinvest.2011.44](https://doi.org/10.1038/labinvest.2011.44)
- Kavanagh K, Wylie AT, Tucker KL et al (2013) Dietary fructose induces endotoxemia and hepatic injury in calorically controlled primates. *Am J Clin Nutr* 98:349–357. doi:[10.3945/ajcn.112.057331](https://doi.org/10.3945/ajcn.112.057331)

- Kawasaki T, Igarashi K, Koeda T et al (2009) Rats fed fructose-enriched diets have characteristics of nonalcoholic hepatic steatosis. *J Nutr* 139:2067–2071. doi:[10.3945/jn.109.105858](https://doi.org/10.3945/jn.109.105858)
- Kelishadi R, Mansourian M, Heidari-Beni M (2014) Association of fructose consumption and components of metabolic syndrome in human studies: a systematic review and meta-analysis. *Nutrition* 30:503–510. doi:[10.1016/j.nut.2013.08.014](https://doi.org/10.1016/j.nut.2013.08.014)
- Koliaki C (2013) Hepatic energy metabolism in human diabetes mellitus, obesity and non-alcoholic fatty liver disease. *Mol Cell Endocrinol* 379:35–42. doi:[10.1016/j.mce.2013.06.002](https://doi.org/10.1016/j.mce.2013.06.002)
- Korieh A, Crouzoulon G (1991) Dietary regulation of fructose metabolism in the intestine and in the liver of the rat. Duration of the effects of a high fructose diet after the return to the standard diet. *Arch Int Physiol Biochim Biophys* 99:455–460
- Lanaspa MA, Cicerchi C, Garcia G et al (2012a) Counteracting roles of AMP deaminase and AMP kinase in the development of fatty liver. *PLoS ONE* 7:e48801. doi:[10.1371/journal.pone.0048801](https://doi.org/10.1371/journal.pone.0048801)
- Lanaspa MA, Sanchez-Lozada LG, Choi YJ et al (2012b) Uric acid induces hepatic steatosis by generation of mitochondrial oxidative stress: Potential role in fructose-dependent and -independent fatty liver. *J Biol Chem* 287:40732–40744. doi:[10.1074/jbc.M112.399899](https://doi.org/10.1074/jbc.M112.399899)
- Lanaspa MA, Ishimoto T, Li N et al (2013) Endogenous fructose production and metabolism in the liver contributes to the development of metabolic syndrome. *Nat Commun* 4:2434. doi:[10.1038/ncomms3434](https://doi.org/10.1038/ncomms3434)
- Latta M, Künstle G, Lucas R et al (2007) ATP-depleting carbohydrates prevent tumor necrosis factor receptor 1-dependent apoptotic and necrotic liver injury in mice. *J Pharmacol Exp Ther* 321:875–883. doi:[10.1124/jpet.107.119958](https://doi.org/10.1124/jpet.107.119958)
- Lettner A, Roden M (2008) Ectopic fat and insulin resistance. *Curr Diab Rep* 8:185–191
- Li YC, Hsieh CC (2014) Lactoferrin dampens high-fructose corn syrup-induced hepatic manifestations of the metabolic syndrome in a murine model. *PLoS ONE* 9:e97341. doi:[10.1371/journal.pone.0097341](https://doi.org/10.1371/journal.pone.0097341)
- Li M, Reynolds CM, Sloboda DM et al (2013) Effects of taurine supplementation on hepatic markers of inflammation and lipid metabolism in mothers and offspring in the setting of maternal obesity. *PLoS ONE* 8:e76961. doi:[10.1371/journal.pone.0076961](https://doi.org/10.1371/journal.pone.0076961)
- Li J-M, Ge C-X, Xu M-X et al (2015) Betaine recovers hypothalamic neural injury by inhibiting astrogliosis and inflammation in fructose-fed rats. *Mol Nutr Food Res* 59:189–202. doi:[10.1002/mnfr.201400307](https://doi.org/10.1002/mnfr.201400307)
- Lim JS, Mietus-Snyder M, Valente A et al (2010) The role of fructose in the pathogenesis of NAFLD and the metabolic syndrome. *Nat Rev Gastroenterol Hepatol* 7:251–264. doi:[10.1038/nrgastro.2010.41](https://doi.org/10.1038/nrgastro.2010.41)
- Lin HZ, Yang SQ, Chuckaree C et al (2000) Metformin reverses fatty liver disease in obese, leptin-deficient mice. *Nat Med* 6:998–1003. doi:[10.1038/79697](https://doi.org/10.1038/79697)
- Lindqvist A, Baelemans A, Erlanson-Albertsson C (2008) Effects of sucrose, glucose and fructose on peripheral and central appetite signals. *Regul Pept* 150:26–32. doi:[10.1016/j.regpep.2008.06.008](https://doi.org/10.1016/j.regpep.2008.06.008)
- Liu J, Zhang H, Ji B et al (2014a) A diet formula of Puerariae radix, Lycium barbarum, Crataegus pinnatifida, and Polygonati rhizoma alleviates insulin resistance and hepatic steatosis in CD-1 mice and HepG2 cells. *Food Funct* 5:1038–1049. doi:[10.1039/c3fo60524h](https://doi.org/10.1039/c3fo60524h)
- Liu X, Xue R, Ji L et al (2014b) Activation of farnesoid X receptor (FXR) protects against fructose-induced liver steatosis via inflammatory inhibition and ADRP reduction. *Biochem Biophys Res Commun* 450:117–123. doi:[10.1016/j.bbrc.2014.05.072](https://doi.org/10.1016/j.bbrc.2014.05.072)
- Lou P-H, Yang G, Huang L et al (2010) Reduced body weight and increased energy expenditure in transgenic mice over-expressing soluble leptin receptor. *PLoS ONE* 5:e11669. doi:[10.1371/journal.pone.0011669](https://doi.org/10.1371/journal.pone.0011669)
- Lustig RH (2010) Fructose: metabolic, hedonic, and societal parallels with ethanol. *J Am Diet Assoc* 110:1307–1321. doi:[10.1016/j.jada.2010.06.008](https://doi.org/10.1016/j.jada.2010.06.008)
- Mahli A, Thasler WE, Patsenker E et al (2015) Identification of cytochrome CYP2E1 as critical mediator of synergistic effects of alcohol and cellular lipid accumulation in hepatocytes in vitro. *Oncotarget* 6:41464–41478. doi:[10.18632/oncotarget.6203](https://doi.org/10.18632/oncotarget.6203)
- Makki K, Taront S, Molendi-Coste O et al (2014) Beneficial metabolic effects of rapamycin are associated with enhanced regulatory cells in diet-induced obese mice. *PLoS ONE* 9:e92684. doi:[10.1371/journal.pone.0092684](https://doi.org/10.1371/journal.pone.0092684)
- Malkusz DC, Yenke I, Rotella FM et al (2015) Dopamine receptor signaling in the medial orbital frontal cortex and the acquisition and expression of fructose-conditioned flavor preferences in rats. *Brain Res* 1596:116–125. doi:[10.1016/j.brainres.2014.11.028](https://doi.org/10.1016/j.brainres.2014.11.028)
- Marriott BP, Cole N, Lee E (2009) National estimates of dietary fructose intake increased from 1977 to 2004 in the United States. *J Nutr* 139:1228S–1235S. doi:[10.3945/jn.108.098277](https://doi.org/10.3945/jn.108.098277)
- Martius G, Alwahsh SM, Rave-Fränk M et al (2014) Hepatic fat accumulation and regulation of FAT/CD36: an effect of hepatic irradiation. *Int J Clin Exp Pathol* 7:5379–5392
- Mate A, De La Hermosa MA, Barfull A et al (2001) Characterization of D-fructose transport by rat kidney brush-border membrane vesicles: changes in hypertensive rats. *Cell Mol Life Sci* 58:1961–1967
- Matz-Soja M, Rennert C, Schönefeld K et al (2016) Hedgehog signaling is a potent regulator of liver lipid metabolism and reveals a GLI-code associated with steatosis. *eLife* 5:e13308. doi:[10.7554/eLife.13308.001](https://doi.org/10.7554/eLife.13308.001)
- Michael MD, Kulkarni RN, Postic C et al (2000) Loss of insulin signaling in hepatocytes leads to severe insulin resistance and progressive hepatic dysfunction. *Mol Cell* 6:87–97
- Montonen J, Järvinen R, Knekt P et al (2007) Consumption of sweetened beverages and intakes of fructose and glucose predict type 2 diabetes occurrence. *J Nutr* 137:1447–1454
- Morishita S, Ono T, Fujisaki C et al (2013) Bovine lactoferrin reduces visceral fat and liver triglycerides in ICR mice. *J Oleo Sci* 62:97–103
- Mortensen OH, Larsen LH, Ørstrup LKH et al (2014) Developmental programming by high fructose decreases phosphorylation efficiency in aging offspring brain mitochondria, correlating with enhanced UCP5 expression. *J Cereb Blood Flow Metab* 34:1205–1211. doi:[10.1038/jcbfm.2014.72](https://doi.org/10.1038/jcbfm.2014.72)
- Mouzaki M, Comelli EM, Arendt BM et al (2013) Intestinal microbiota in patients with nonalcoholic fatty liver disease. *Hepatology* 58:120–127. doi:[10.1002/hep.26319](https://doi.org/10.1002/hep.26319)
- Mukai Y, Ozaki H, Serita Y, Sato S (2014) Maternal fructose intake during pregnancy modulates hepatic and hypothalamic AMP-activated protein kinase signalling in a sex-specific manner in offspring. *Clin Exp Pharmacol Physiol* 41:331–337. doi:[10.1111/1440-1681.12225](https://doi.org/10.1111/1440-1681.12225)
- Nair S, Chacko VP, Arnold C, Diehl AM (2003) Hepatic ATP reserve and efficiency of replenishing: comparison between obese and nonobese normal individuals. *Am J Gastroenterol* 98:466–470. doi:[10.1016/S0002-9270\(02\)05889-6](https://doi.org/10.1016/S0002-9270(02)05889-6)
- Nair S, Diehl AM, Wiseman M et al (2004) Metformin in the treatment of non-alcoholic steatohepatitis: a pilot open label trial. *Aliment Pharmacol Ther* 20:23–28. doi:[10.1111/j.1365-2036.2004.02025.x](https://doi.org/10.1111/j.1365-2036.2004.02025.x)

- Nakagawa T, Hu H, Zharikov S et al (2006) A causal role for uric acid in fructose-induced metabolic syndrome. *Am J Physiol Renal Physiol* 290:F625–F631. doi:[10.1152/ajprenal.00140.2005](https://doi.org/10.1152/ajprenal.00140.2005)
- Nan Y, Wang R, Fu N (2014) Peroxisome proliferator-activated receptor α , a potential therapeutic target for alcoholic liver disease. *World J Gastroenterol* 20:8055–8060. doi:[10.3748/wjg.v20.i25.8055](https://doi.org/10.3748/wjg.v20.i25.8055)
- Narasimhan A, Chinnaiyan M, Karundevi B (2015) Ferulic acid regulates hepatic GLUT2 gene expression in high fat and fructose-induced type-2 diabetic adult male rat. *Eur J Pharmacol* 761:391–397. doi:[10.1016/j.ejphar.2015.04.043](https://doi.org/10.1016/j.ejphar.2015.04.043)
- Niklas J, Bonin A, Mangin S et al (2012) Central energy metabolism remains robust in acute steatotic hepatocytes challenged by a high free fatty acid load. *BMB Rep* 45:396–401
- Okada E, Oida K, Tada H et al (1999) Hyperhomocysteinemia is a risk factor for coronary arteriosclerosis in Japanese patients with type 2 diabetes. *Diabetes Care* 22:484–490. doi:[10.2337/diacare.22.3.484](https://doi.org/10.2337/diacare.22.3.484)
- Onyesom I (2005) Honey-induced stimulation of blood ethanol elimination and its influence on serum triacylglycerol and blood pressure in man. *Ann Nutr Metab* 49:319–324. doi:[10.1159/000087336](https://doi.org/10.1159/000087336)
- Oron-Herman M, Rosenthal T, Sela B-A (2003) Hyperhomocysteinemia as a component of syndrome X. *Metabolism* 52:1491–1495
- Ouyang X, Cirillo P, Sautin Y et al (2008) Fructose consumption as a risk factor for non-alcoholic fatty liver disease. *J Hepatol* 48:993–999. doi:[10.1016/j.jhep.2008.02.011](https://doi.org/10.1016/j.jhep.2008.02.011)
- Page KA, Chan O, Arora J et al (2013) Effects of fructose vs glucose on regional cerebral blood flow in brain regions involved with appetite and reward pathways. *JAMA* 309:63. doi:[10.1001/jama.2012.116975](https://doi.org/10.1001/jama.2012.116975)
- Pastoret A, Marcos R, Sampayo-Reyes A et al (2013) Inhibition of hepatocyte nuclear factor 1 and 4 alpha (HNF1 α and HNF4 α) as a mechanism of arsenic carcinogenesis. *Arch Toxicol* 87:1001–1012. doi:[10.1007/s00204-012-0948-6](https://doi.org/10.1007/s00204-012-0948-6)
- Pessayre D, Fromenty B (2005) NASH: a mitochondrial disease. *J Hepatol* 42:928–940. doi:[10.1016/j.jhep.2005.03.004](https://doi.org/10.1016/j.jhep.2005.03.004)
- Petrie JL, Patman GL, Sinha I et al (2013) The rate of production of uric acid by hepatocytes is a sensitive index of compromised cell ATP homeostasis. *Am J Physiol Endocrinol Metab* 305:E1255–E1265. doi:[10.1152/ajpendo.00214.2013](https://doi.org/10.1152/ajpendo.00214.2013)
- Petta S, Marchesini G, Caracausi L et al (2013) Industrial, not fruit fructose intake is associated with the severity of liver fibrosis in genotype 1 chronic hepatitis C patients. *J Hepatol* 59:1169–1176. doi:[10.1016/j.jhep.2013.07.037](https://doi.org/10.1016/j.jhep.2013.07.037)
- Prakash P, Singh V, Jain M et al (2014) Silymarin ameliorates fructose induced insulin resistance syndrome by reducing de novo hepatic lipogenesis in the rat. *Eur J Pharmacol* 727:15–28. doi:[10.1016/j.ejphar.2014.01.038](https://doi.org/10.1016/j.ejphar.2014.01.038)
- Pyo Y-H, Lee K-W (2014) Preventive effect of Monascus-fermented products enriched with ubiquinones on type 2 diabetic rats induced by a high-fructose plus high-fat diet. *J Med Food* 17:826–829. doi:[10.1089/jmf.2013.3001](https://doi.org/10.1089/jmf.2013.3001)
- Qu H, Li J, Chen W et al (2014) Differential expression of the melanocortin-4 receptor in male and female C57BL/6J mice. *Mol Biol Rep* 41:3245–3256. doi:[10.1007/s11033-014-3187-5](https://doi.org/10.1007/s11033-014-3187-5)
- Raatz SK, Johnson LK, Picklo MJ (2015) Consumption of honey, sucrose, and high-fructose corn syrup produces similar metabolic effects in glucose-tolerant and -intolerant individuals. *J Nutr* 145:2265–2272. doi:[10.3945/jn.115.218016](https://doi.org/10.3945/jn.115.218016)
- Rashid A, Wu T-C, Huang C et al (1999) Mitochondrial proteins that regulate apoptosis and necrosis are induced in mouse fatty liver. *Hepatology* 29:1131–1138
- Rebollo A, Roglans N, Baena M et al (2014) Liquid fructose downregulates Sirt1 expression and activity and impairs the oxidation of fatty acids in rat and human liver cells. *Biochim Biophys Acta Mol Cell Biol Lipids* 1841:514–524. doi:[10.1016/j.bbalip.2014.01.002](https://doi.org/10.1016/j.bbalip.2014.01.002)
- Ren L-P, Chan SMH, Zeng X-Y et al (2012) Differing endoplasmic reticulum stress response to excess lipogenesis versus lipid oversupply in relation to hepatic steatosis and insulin resistance. *PLoS ONE* 7:e30816. doi:[10.1371/journal.pone.0030816](https://doi.org/10.1371/journal.pone.0030816)
- Ren D, Zhao Y, Nie Y et al (2014) Chemical composition of *Pleurotus eryngii* polysaccharides and their inhibitory effects on high-fructose diet-induced insulin resistance and oxidative stress in mice. *Food Funct* 5:2609–2620. doi:[10.1039/c3fo60640f](https://doi.org/10.1039/c3fo60640f)
- Renaud HJ, Cui JY, Lu H, Klaassen CD (2014) Effect of diet on expression of genes involved in lipid metabolism, oxidative stress, and inflammation in mouse liver—insights into mechanisms of hepatic steatosis. *PLoS ONE* 9:e88584. doi:[10.1371/journal.pone.0088584](https://doi.org/10.1371/journal.pone.0088584)
- Riby JE, Fujisawa T, Kretchmer N (1993) Fructose absorption. *Am J Clin Nutr* 58:748S–753S
- Ritze Y, Bárdos G, D’Haese JG et al (2014) Effect of high sugar intake on glucose transporter and weight regulating hormones in mice and humans. *PLoS ONE* 9:e101702. doi:[10.1371/journal.pone.0101702](https://doi.org/10.1371/journal.pone.0101702)
- Rivera FP, Medina AM, Bezada S et al (2013) Bovine lactoferrin decreases cholera-toxin-induced intestinal fluid accumulation in mice by ganglioside interaction. *PLoS ONE* 8:e59253. doi:[10.1371/journal.pone.0059253](https://doi.org/10.1371/journal.pone.0059253)
- Rodrigues DF, do Carmo Henriques MC, Oliveira MC et al (2014) Acute intake of a high-fructose diet alters the balance of adipokine concentrations and induces neutrophil influx in the liver. *J Nutr Biochem* 25:388–394. doi:[10.1016/j.jnutbio.2013.11.012](https://doi.org/10.1016/j.jnutbio.2013.11.012)
- Rokutan M, Yabe D, Komoto I et al (2015) A case of insulinoma with non-alcoholic fatty liver disease: roles of hyperphagia and hyperinsulinemia in pathogenesis of the disease. *Endocr J* 62:1025–1030. doi:[10.1507/endocrj.EJ14-0590](https://doi.org/10.1507/endocrj.EJ14-0590)
- Rorabaugh JM, Stratford JM, Zahniser NR (2014) A relationship between reduced nucleus accumbens shell and enhanced lateral hypothalamic orexin neuronal activation in long-term fructose bingeing behavior. *PLoS ONE* 9:e95019. doi:[10.1371/journal.pone.0095019](https://doi.org/10.1371/journal.pone.0095019)
- Rovenko BM, Perkhulyn NV, Gospodaryov DV et al (2015) High consumption of fructose rather than glucose promotes a diet-induced obese phenotype in *Drosophila melanogaster*. *Comp Biochem Physiol Part A Mol Integr Physiol* 180:75–85. doi:[10.1016/j.cbpa.2014.11.008](https://doi.org/10.1016/j.cbpa.2014.11.008)
- Sakar Y, Nazaret C, Lettéron P et al (2009) Positive regulatory control loop between gut leptin and intestinal GLUT2/GLUT5 transporters links to hepatic metabolic functions in rodents. *PLoS ONE* 4:e7935. doi:[10.1371/journal.pone.0007935](https://doi.org/10.1371/journal.pone.0007935)
- Samuel VT, Liu Z-X, Qu X et al (2004) Mechanism of hepatic insulin resistance in non-alcoholic fatty liver disease. *J Biol Chem* 279:32345–32353. doi:[10.1074/jbc.M313478200](https://doi.org/10.1074/jbc.M313478200)
- Sapp V, Gaffney L, EauClaire SF, Matthews RP (2014) Fructose leads to hepatic steatosis in zebrafish that is reversed by mTOR inhibition. *Hepatology* 60:1581–1592. doi:[10.1002/hep.27284](https://doi.org/10.1002/hep.27284)
- Saravanan M, Pandikumar P, Saravanan S et al (2014) Lipolytic and antiadipogenic effects of (3,3-dimethylallyl) halfordinol on 3T3-L1 adipocytes and high fat and fructose diet induced obese C57/BL6J mice. *Eur J Pharmacol* 740:714–721. doi:[10.1016/j.ejphar.2014.06.004](https://doi.org/10.1016/j.ejphar.2014.06.004)
- Scholz R, Nohl H (1976) Mechanism of the stimulatory effect of fructose on ethanol oxidation in perfused rat liver. *Eur J Biochem* 63:449–458
- Schwarz J-M, Noworolski SM, Wen MJ et al (2015) Effect of a high-fructose weight-maintaining diet on lipogenesis and liver fat. *J Clin Endocrinol Metab* 100:2434–2442. doi:[10.1210/jc.2014-3678](https://doi.org/10.1210/jc.2014-3678)

- Sclafani A, Ackroff K (1994) Glucose- and fructose-conditioned flavor preferences in rats: taste versus postingestive conditioning. *Physiol Behav* 56:399–405
- Sharma S, Mells JE, Fu PP et al (2011) GLP-1 analogs reduce hepatocyte steatosis and improve survival by enhancing the unfolded protein response and promoting macroautophagy. *PLoS ONE* 6:e25269. doi:[10.1371/journal.pone.0025269](https://doi.org/10.1371/journal.pone.0025269)
- Shawky NM, Shehatou GSG, Abdel Rahim M et al (2014) Levocetirizine ameliorates high fructose diet-induced insulin resistance, vascular dysfunction and hepatic steatosis in rats. *Eur J Pharmacol* 740:353–363. doi:[10.1016/j.ejphar.2014.07.021](https://doi.org/10.1016/j.ejphar.2014.07.021)
- Shi P, Chen B, Chen C et al (2015) Honey reduces blood alcohol concentration but not affects the level of serum MDA and GSH-Px activity in intoxicated male mice models. *BMC Complement Altern Med* 15:225. doi:[10.1186/s12906-015-0766-5](https://doi.org/10.1186/s12906-015-0766-5)
- Shulga N, Pastorino JG (2014) Mitoneet mediates TNF α -induced necroptosis promoted by exposure to fructose and ethanol. *J Cell Sci* 127:896–907. doi:[10.1242/jcs.140764](https://doi.org/10.1242/jcs.140764)
- Sloboda DM, Li M, Patel R et al (2014) Early life exposure to fructose and offspring phenotype: implications for long term metabolic homeostasis. *J Obes* 2014:1–10. doi:[10.1155/2014/203474](https://doi.org/10.1155/2014/203474)
- Soleimani M (2011) Dietary fructose, salt absorption and hypertension in metabolic syndrome: towards a new paradigm. *Acta Physiol* 201:55–62. doi:[10.1111/j.1748-1716.2010.02167.x](https://doi.org/10.1111/j.1748-1716.2010.02167.x)
- Song M, Schuschke DA, Zhou Z et al (2012) High fructose feeding induces copper deficiency in Sprague-Dawley rats: a novel mechanism for obesity related fatty liver. *J Hepatol* 56:433–440. doi:[10.1016/j.surg.2006.10.010](https://doi.org/10.1016/j.surg.2006.10.010). Use
- Song M, Chen T, Prough RA et al (2016) Chronic alcohol consumption causes liver injury in high-fructose-fed male mice through enhanced hepatic inflammatory response. *Alcohol Clin Exp Res* 40:518–528. doi:[10.1111/acer.12994](https://doi.org/10.1111/acer.12994)
- Speicher T, Köhler UA, Choukèr A et al (2012) Fructose protects murine hepatocytes from tumor necrosis factor-induced apoptosis by modulating JNK signaling. *J Biol Chem* 287:1837–1846. doi:[10.1074/jbc.M111.266742](https://doi.org/10.1074/jbc.M111.266742)
- Spruss A, Kanuri G, Wagnerberger S et al (2009) Toll-like receptor 4 is involved in the development of fructose-induced hepatic steatosis in mice. *Hepatology* 50:1094–1104. doi:[10.1002/hep.23122](https://doi.org/10.1002/hep.23122)
- Spruss A, Kanuri G, Uebel K et al (2011) Role of the inducible nitric oxide synthase in the onset of fructose-induced steatosis in mice. *Antioxid Redox Signal* 14:2121–2135. doi:[10.1089/ars.2010.3263](https://doi.org/10.1089/ars.2010.3263)
- Spruss A, Henkel J, Kanuri G et al (2012a) Female mice are more susceptible to nonalcoholic fatty liver disease: sex-specific regulation of the hepatic AMP-activated protein kinase-plasminogen activator inhibitor 1 cascade, but not the hepatic endotoxin response. *Mol Med* 18:1346–1355. doi:[10.2119/molmed.2012.00223](https://doi.org/10.2119/molmed.2012.00223)
- Spruss A, Kanuri G, Stahl C et al (2012b) Metformin protects against the development of fructose-induced steatosis in mice: role of the intestinal barrier function. *Lab Invest* 92:1020–1032. doi:[10.1038/labinvest.2012.75](https://doi.org/10.1038/labinvest.2012.75)
- Stanhope KL, Schwarz JM, Keim NL et al (2009) Consuming fructose-sweetened, not glucose-sweetened, beverages increases visceral adiposity and lipids and decreases insulin sensitivity in overweight/obese humans. *J Clin Invest* 119:1322–1334. doi:[10.1172/JCI37385](https://doi.org/10.1172/JCI37385)
- Suwannaphet W, Meeprom A, Yibchok-Anun S, Adisakwattana S (2010) Preventive effect of grape seed extract against high-fructose diet-induced insulin resistance and oxidative stress in rats. *Food Chem Toxicol* 48:1853–1857. doi:[10.1016/j.fct.2010.04.021](https://doi.org/10.1016/j.fct.2010.04.021)
- Tappy L, Egli L, Lecoultré V, Schneider P (2013) Effects of fructose-containing caloric sweeteners on resting energy expenditure and energy efficiency: a review of human trials. *Nutr Metab* 10:54. doi:[10.1186/1743-7075-10-54](https://doi.org/10.1186/1743-7075-10-54)
- Thuy S, Ladurner R, Volynets V et al (2008) Nonalcoholic fatty liver disease in humans is associated with increased plasma endotoxin and plasminogen activator inhibitor 1 concentrations and with fructose intake. *J Nutr* 138:1452–1455
- Tilg H, Moschen AR, Roden M (2016) NAFLD and diabetes mellitus. *Nat Rev Gastroenterol Hepatol*. doi:[10.1038/nrgastro.2016.147](https://doi.org/10.1038/nrgastro.2016.147)
- Tiniakos DG, Vos MB, Brunt EM (2010) Nonalcoholic fatty liver disease: pathology and pathogenesis. *Annu Rev Pathol* 5:145–171. doi:[10.1146/annurev-pathol-121808-102132](https://doi.org/10.1146/annurev-pathol-121808-102132)
- Tomita K, Tamiya G, Ando S et al (2006) Tumour necrosis factor α signalling through activation of Kupffer cells plays an essential role in liver fibrosis of non-alcoholic steatohepatitis in mice. *Gut* 55:415–424. doi:[10.1136/gut.2005.071118](https://doi.org/10.1136/gut.2005.071118)
- Uzuegbu UE, Onyesom I (2009) Fructose-induced increase in ethanol metabolism and the risk of Syndrome X in man. *C R Biol* 332:534–538. doi:[10.1016/j.crvi.2009.01.007](https://doi.org/10.1016/j.crvi.2009.01.007)
- Vilà L, Rebollo A, Adalsteisson GS et al (2011) Reduction of liver fructokinase expression and improved hepatic inflammation and metabolism in liquid fructose-fed rats after atorvastatin treatment. *Toxicol Appl Pharmacol* 251:32–40. doi:[10.1016/j.taap.2010.11.011](https://doi.org/10.1016/j.taap.2010.11.011)
- Volynets V, Spruss A, Kanuri G et al (2010) Protective effect of bile acids on the onset of fructose-induced hepatic steatosis in mice. *J Lipid Res* 51:3414–3424. doi:[10.1194/jlr.M007179](https://doi.org/10.1194/jlr.M007179)
- Volynets V, Machann J, Küper MA et al (2013) A moderate weight reduction through dietary intervention decreases hepatic fat content in patients with non-alcoholic fatty liver disease (NAFLD): a pilot study. *Eur J Nutr* 52:527–535. doi:[10.1007/s00394-012-0355-z](https://doi.org/10.1007/s00394-012-0355-z)
- Vos MB, Kimmons JE, Gillespie C et al (2008) Dietary fructose consumption among US children and adults: the Third National Health and Nutrition Examination Survey. *Medscape J Med* 10:160
- Weiland T, Klein K, Zimmermann M et al (2012) Selective protection of human liver tissue in TNF-targeting of cancers of the liver by transient depletion of adenosine triphosphate. *PLoS ONE* 7:e52496. doi:[10.1371/journal.pone.0052496](https://doi.org/10.1371/journal.pone.0052496)
- Welsh JA, Sharma AJ, Grellinger L, Vos MB (2011) Consumption of added sugars is decreasing in the United States. *Am J Clin Nutr* 94:726–734. doi:[10.3945/ajcn.111.018366](https://doi.org/10.3945/ajcn.111.018366)
- White JS (2008) Straight talk about high-fructose corn syrup: what it is and what it. *Am J Clin Nutr* 88:1716S–1721S. doi:[10.3945/ajcn.2008.25825B.2](https://doi.org/10.3945/ajcn.2008.25825B.2)
- Wigg AJ, Roberts-Thomson IC, Dymock RB et al (2001) The role of small intestinal bacterial overgrowth, intestinal permeability, endotoxaemia, and tumour necrosis factor α in the pathogenesis of non-alcoholic steatohepatitis. *Gut* 48:206–211
- Xu M, Alwahsh SM, Ramadori G et al (2013) Upregulation of hepatic melanocortin 4 receptor during rat liver regeneration. *J Surg Res*. doi:[10.1016/j.jss.2013.12.019](https://doi.org/10.1016/j.jss.2013.12.019)
- Xu T, Zheng L, Xu L et al (2014) Protective effects of dioscin against alcohol-induced liver injury. *Arch Toxicol*. doi:[10.1007/s00204-013-1148-8](https://doi.org/10.1007/s00204-013-1148-8)
- Xu M-X, Yu R, Shao L-F et al (2016) Up-regulated fractalkine (FKN) and its receptor CX3CR1 are involved in fructose-induced neuroinflammation: suppression by curcumin. *Brain Behav Immun* 58:69–81. doi:[10.1016/j.bbi.2016.01.001](https://doi.org/10.1016/j.bbi.2016.01.001)
- Yin Q, Ma Y, Hong Y et al (2014) Lycopene attenuates insulin signaling deficits, oxidative stress, neuroinflammation, and cognitive impairment in fructose-drinking insulin resistant rats. *Neuropharmacology* 86:389–396. doi:[10.1016/j.neuropharm.2014.07.020](https://doi.org/10.1016/j.neuropharm.2014.07.020)
- Yki-Järvinen H (2010) Nutritional modulation of nonalcoholic fatty liver disease and insulin resistance: human data. *Curr*

- Opin Clin Nutr Metab Care 13:709–714. doi:[10.1097/MCO.0b013e32833f4b34](https://doi.org/10.1097/MCO.0b013e32833f4b34)
- Zhao Y, Yang X, Ren D et al (2014) Preventive effects of jujube polysaccharides on fructose-induced insulin resistance and dyslipidemia in mice. Food Funct 5:1771. doi:[10.1039/C3FO60707K](https://doi.org/10.1039/C3FO60707K)
- Zheng J, Xiao X, Zhang Q et al (2015) Maternal and post-weaning high-fat, high-sucrose diet modulates glucose homeostasis and hypothalamic POMC promoter methylation in mouse offspring. Metab Brain Dis 30:1129–1137. doi:[10.1007/s11011-015-9678-9](https://doi.org/10.1007/s11011-015-9678-9)
- Zhong W, Zhang W, Li Q et al (2015) Pharmacological activation of aldehyde dehydrogenase 2 by Alda-1 reverses alcohol-induced hepatic steatosis and cell death in mice. J Hepatol 62:1375–1381. doi:[10.1016/j.jhep.2014.12.022](https://doi.org/10.1016/j.jhep.2014.12.022)